

ELEC3130 Spring 2026

Digital Image Processing

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The Hong Kong University of Science and Technology
Department of Electronic and Computer Engineering

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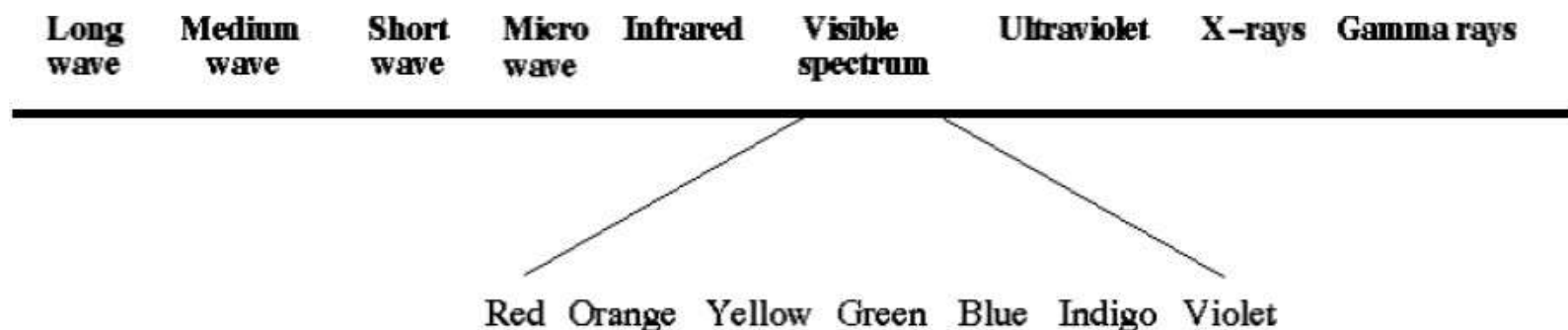
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- ▶ TA Hour: Wed
1:00pm-2:00pm
- ▶ Room: 4218D

Overview

- ▶ What is Digital Image?
- ▶ What is Processing on Images?
- ▶ Why is it important?
- ▶ What knowledge do you need in order to take this course?
- ▶ What is going to be covered in this course?
- ▶ What textbooks are we going to use?
- ▶ Where to obtain additional information?
- ▶ How are we going to teach this course?

What are Images?

- ▶ (Sets of) Objects perceived by Human Visual System (HVS)
- ▶ Light waves Emitted or Reflected from Objects are Received and Processed by Eye and Brain.
- ▶ Light waves are Electro Magnetic waves



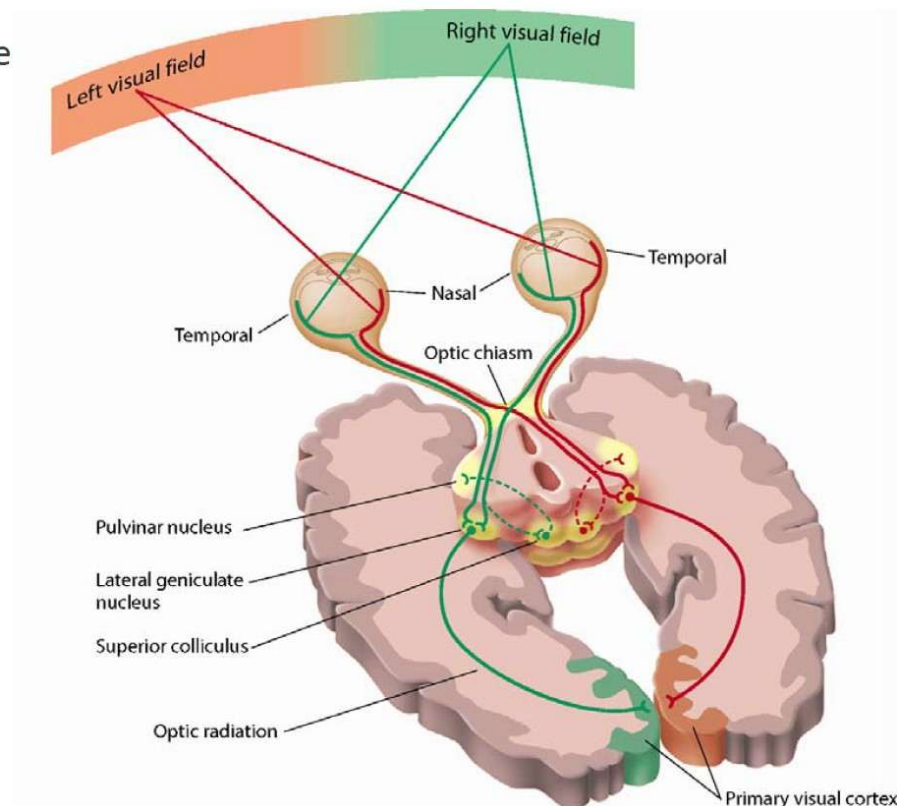
What is visual system?

Vision

Visual information dominates perceptions and frame the way we think.

Vision pathway

- Photoreceptors
 - Retina: cones (color) and rods (night vision)
- Optic nerve
 - Ganglion cells, compress the information
- Lateral geniculate nucleus (LGN)
- Primary visual cortex (V1)



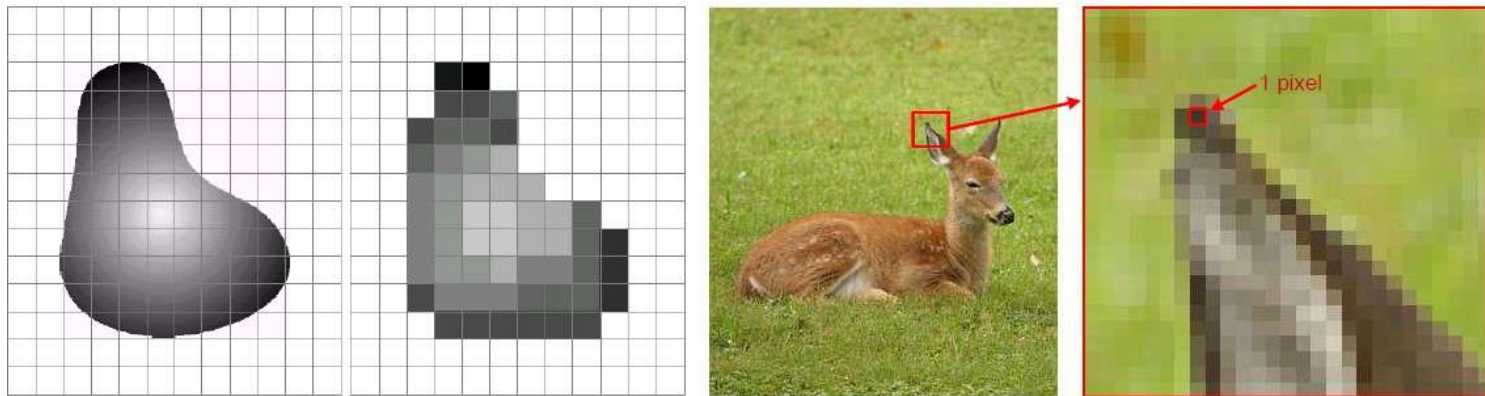
What are Images?



- ▶ Image: a visual representation in form of a function $f(x, y)$ where f is related to the brightness (or color) at point (x, y)
- ▶ Most images are defined over a rectangle
- ▶ Continuous in amplitude and space

What is Digital Image?

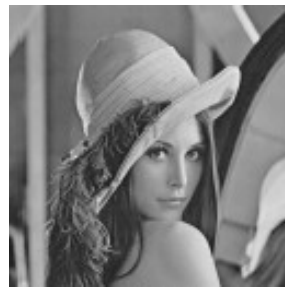
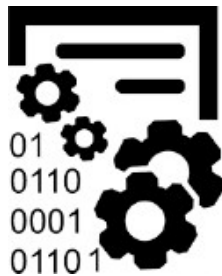
- ▶ Discrete samples $f[x, y]$ representing continuous image $f(x, y)$. f , x , and y are all finite, discrete quantities
- ▶ Each element of the 2-d array $f[x, y]$ is called a pixel
- ▶ Remember digitization implies that a digital image is an approximation of a real scene



- ▶ Knowledge in ELEC2100 (signals and systems)/3100 (signal processing and communications) will be helpful

Basic Types of Digital Images

- ▶ Binary: (Black and White) Only two kinds of pixels
- ▶ Gray Scale: Several levels of gray (Ex. 8 bit gray scale has 256 different gray levels)
- ▶ Color: Usually 3 components
- ▶ Half-tone: Gives effect of gray scale using only black and white pixels

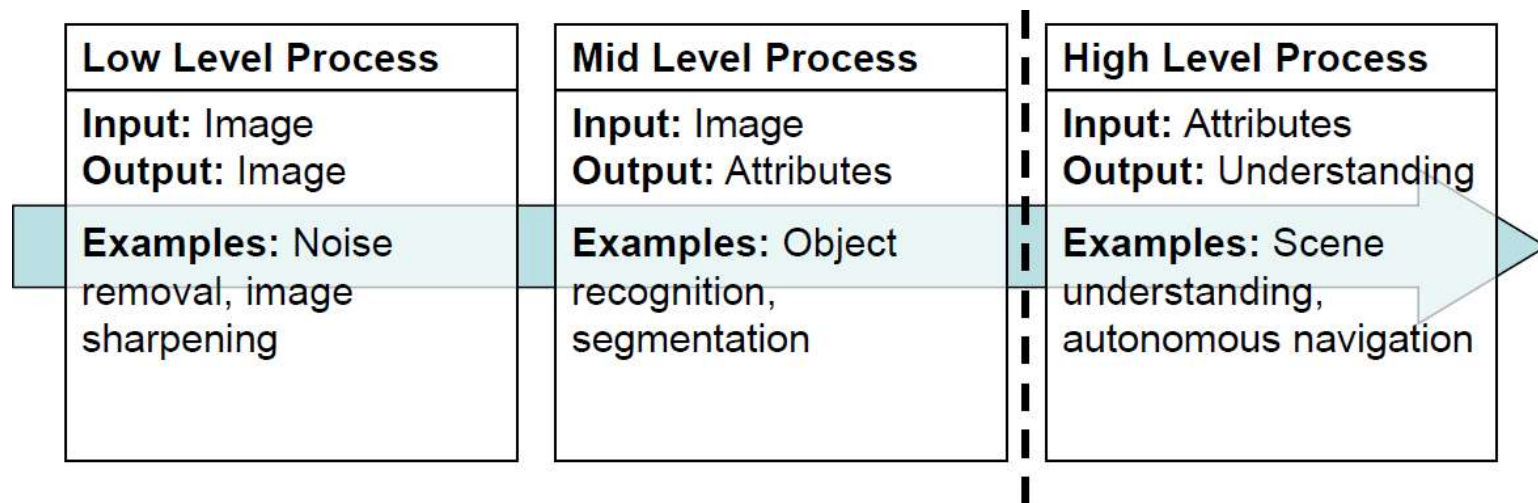


What is Processing on Digital Image ?

- ▶ Focuses on two major tasks
 - ▶ Improvement of pictorial information for human interpretation
 - ▶ Processing of image data for storage, transmission and extraction of pictorial information
- ▶ Some argument about where image processing ends and other fields, such as image analysis and computer vision, start

What is Processing on Digital Image? (cont.)

- ▶ The continuum from image processing to computer vision can be broken up into low-, mid- and high-level processes



In this course we will
stop here

Importance of Digital Image

- ▶ Visual perception is the most important source of information for human being
 - ▶ Seeing it once is more than hearing it hundred times
 - ▶ I see (not I hear nor I touch)
- ▶ Spatial structure information in the perception
- ▶ Digital image has shown many important applications

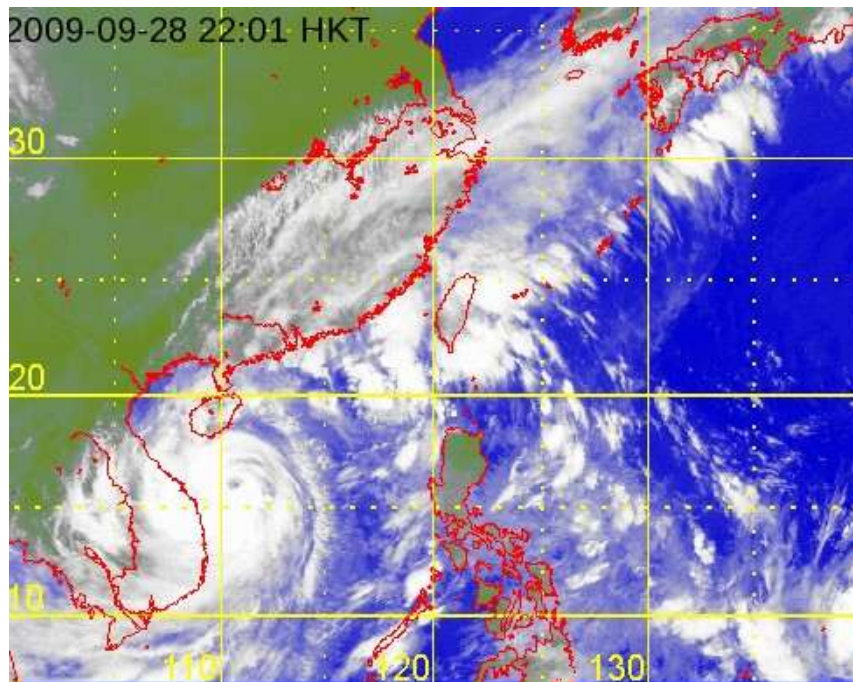
Why processing digital images?

- ▶ Acquire an image!
 - ▶ Correct aperture and color balance
 - ▶ Reconstruct image from projections
- ▶ Prepare for display or printing!
 - ▶ Adjust image size
 - ▶ Color mapping, gamma-correction, halftoning
- ▶ Facilitate picture storage and transmission!
 - ▶ Efficiently store an image in a digital camera
 - ▶ Send an image from space
- ▶ Enhance and restore images!
 - ▶ Touch up personal photos
 - ▶ Color enhancement for security screening
- ▶ Extract information from images!
 - ▶ Read 2-d bar codes
 - ▶ Character recognition



Examples of Images

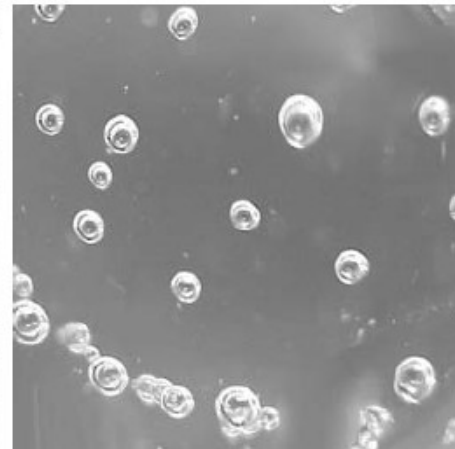
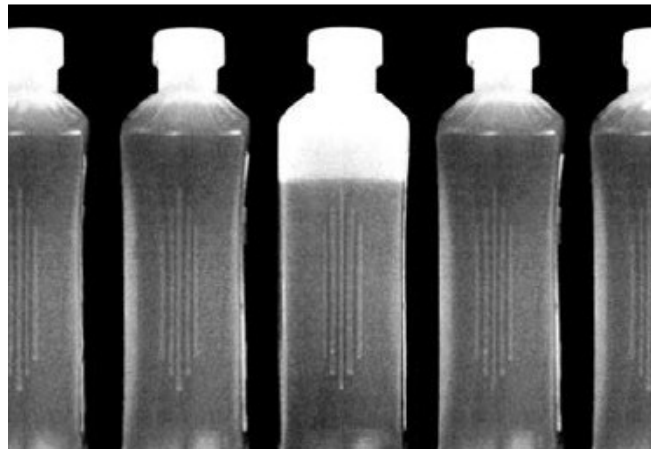
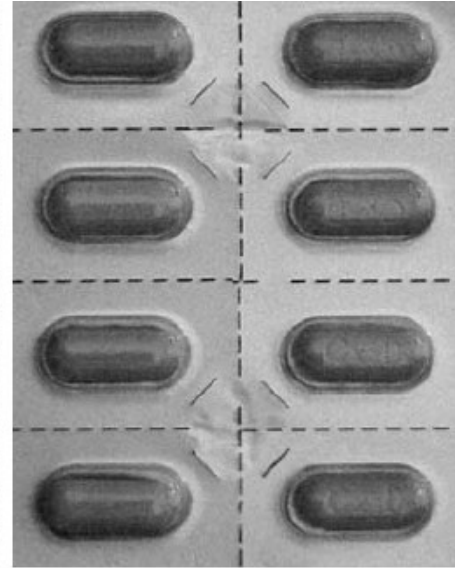
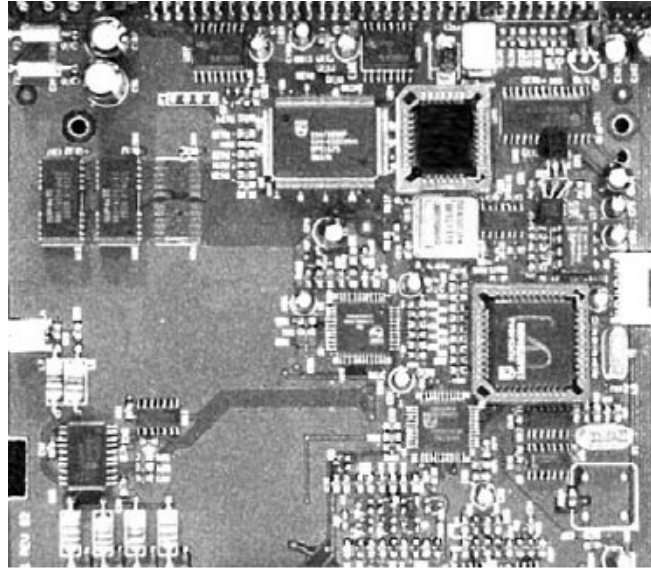
- ▶ Remote sensing and weather forecast



- ▶ Law Enforcement

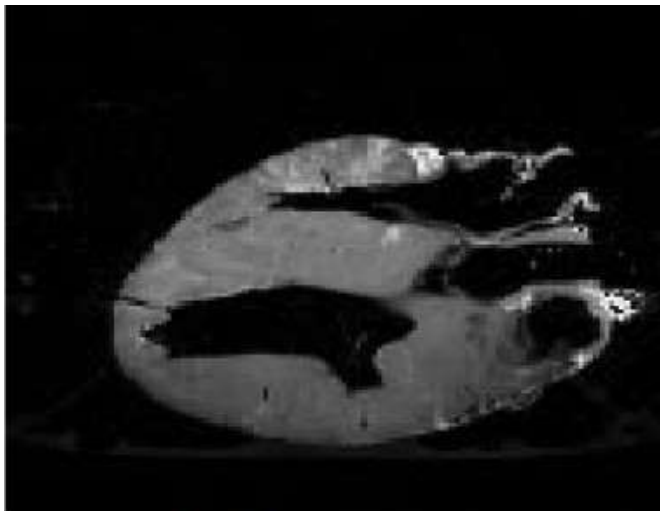


Examples: Visual inspection of manufactured goods

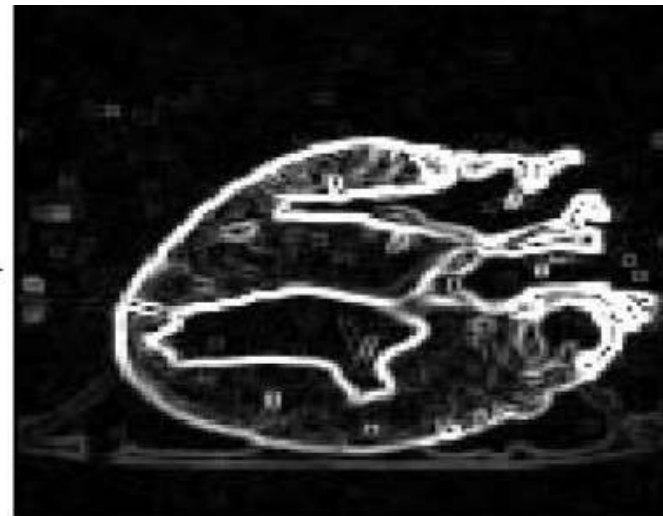


Examples: Medical Imaging

- ▶ Take slice from MRI scan (in radio band) of canine heart, and find boundaries between types of tissue
 - ▶ Image with gray levels representing tissue density
 - ▶ Use a suitable filter to highlight edges



Original MRI Image of a Dog Heart



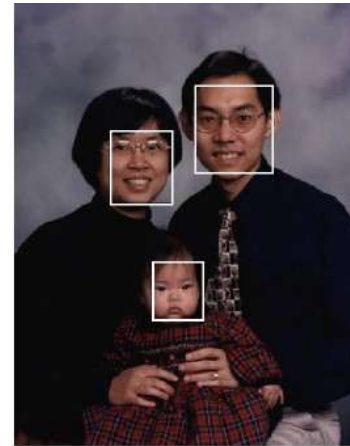
Edge Detection Image

Examples: Artistic Effects

- ▶ Artistic effects are used to make images more visually appealing, to add special effects and to make composite images



Examples: Face detection



source: Henry Chang, Ulises Robles, EE368 class project, spring 2000.

Examples: Face Blurring



source: Henry Chang, Unises Kovles, EE300 class project, spring 2000.

Examples: Paint recognition

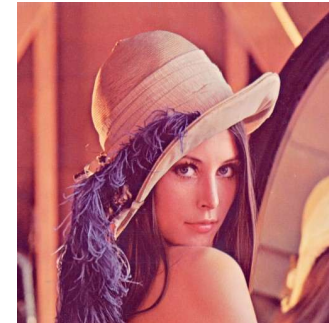


source: Henry Chang, Onses Robles, EE508 class project, spring 2000.

Examples: CD recognition

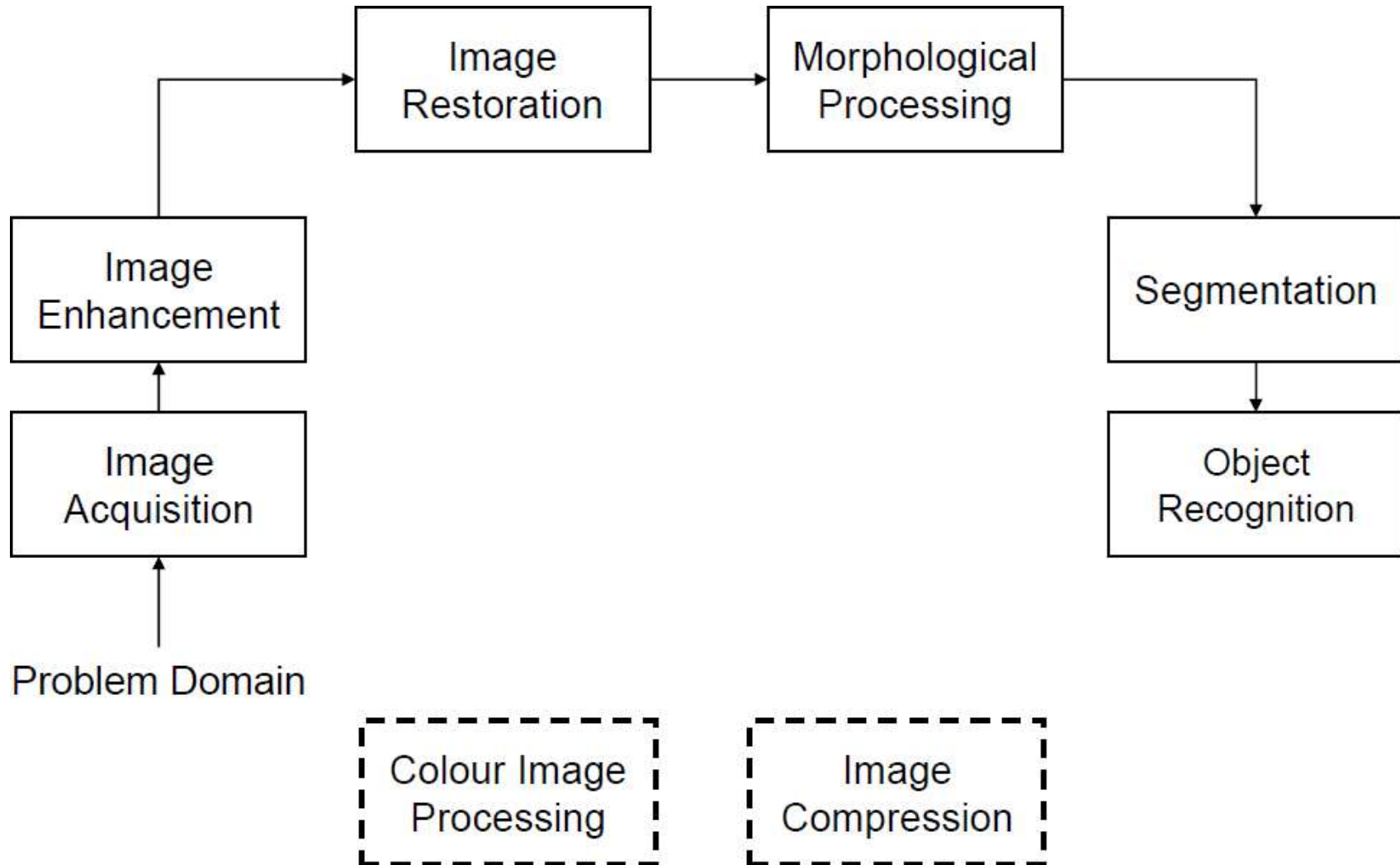


Image $\neq$ Information

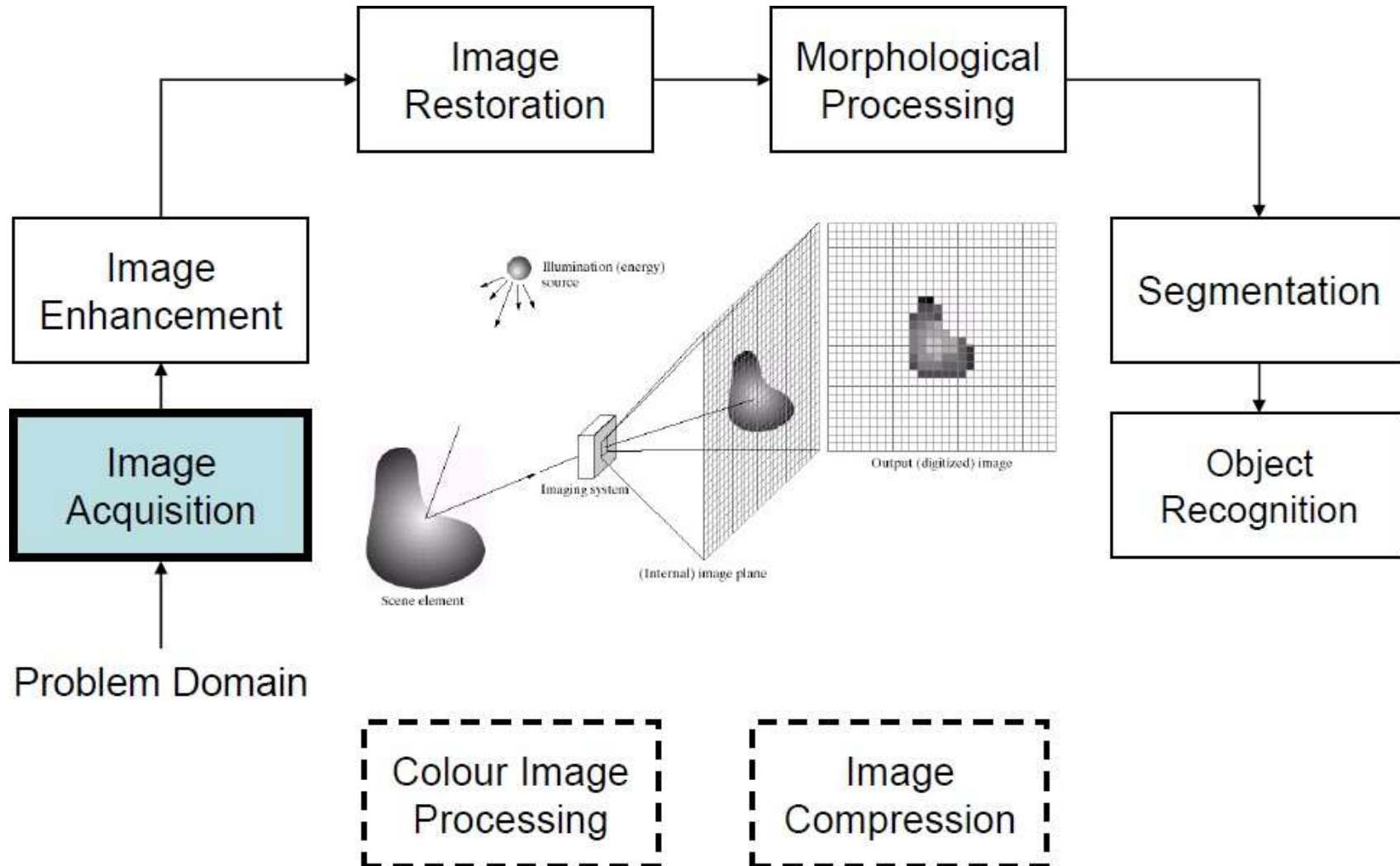


- ▶ Same information in the above two images?
- ▶ How to measure the information quantitatively?
- ▶ What processing do we need to extract information from images?
- ▶ Interesting story at <http://www.lenna.org/>

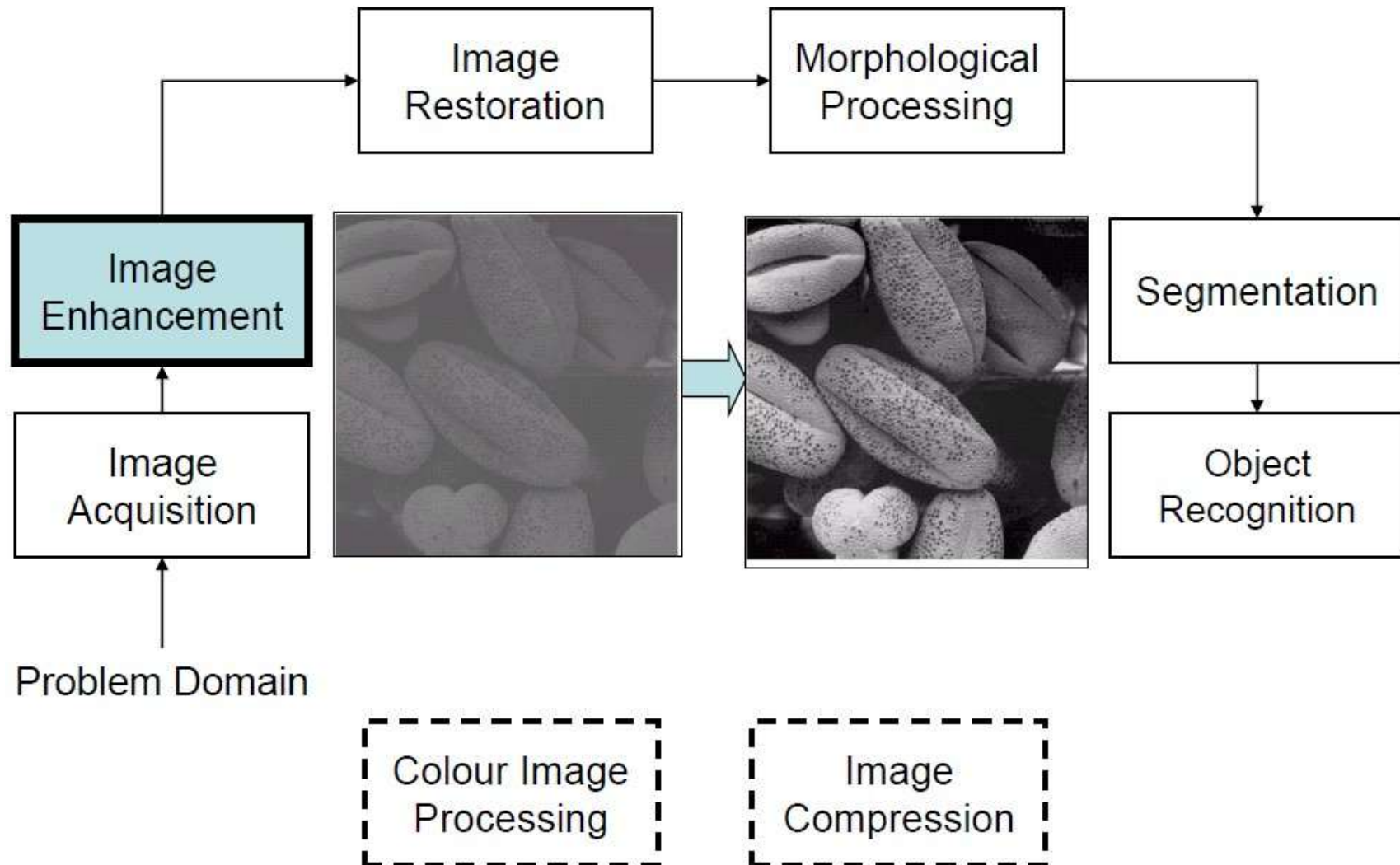
Key stages



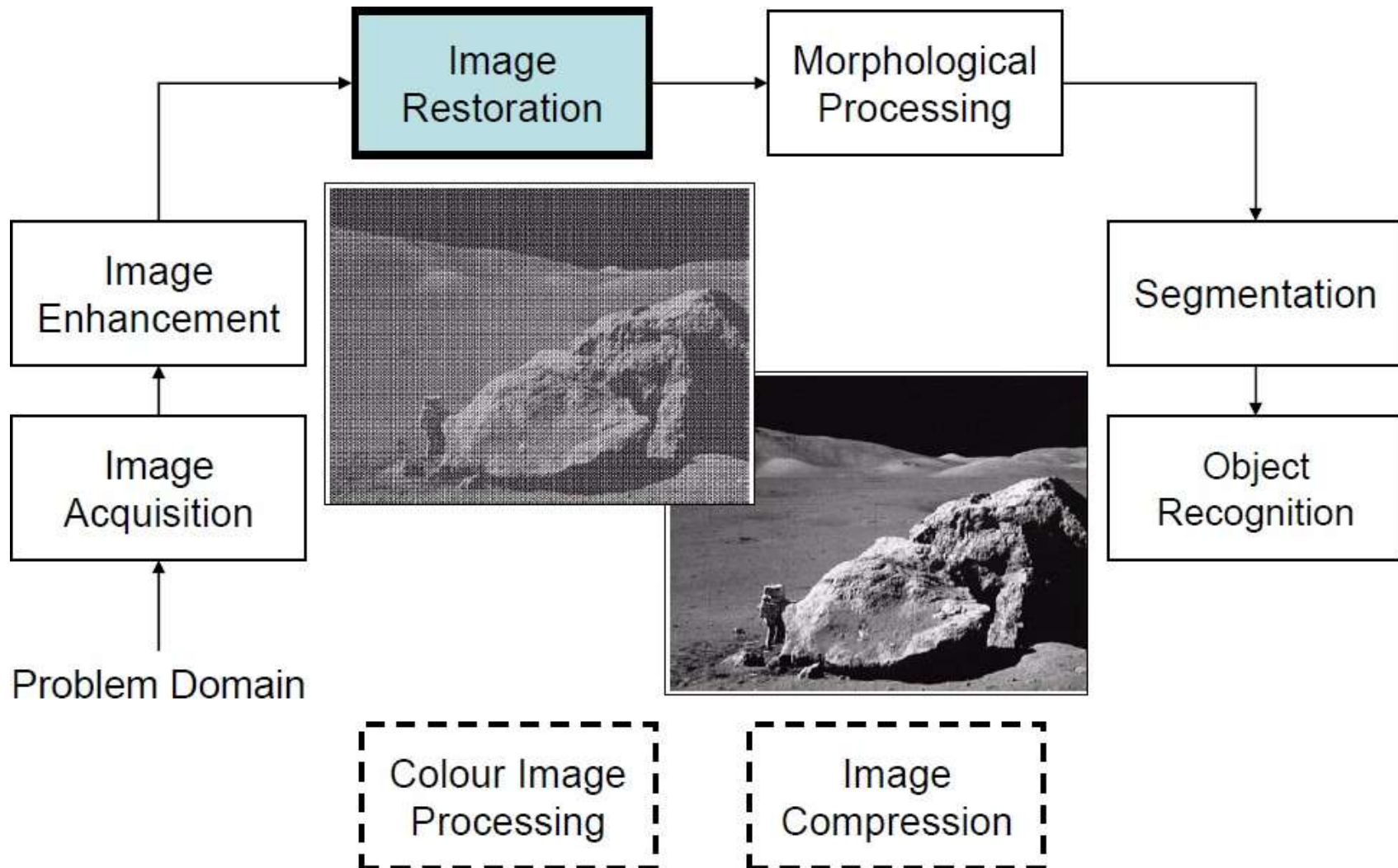
Key stages: Image Acquisition



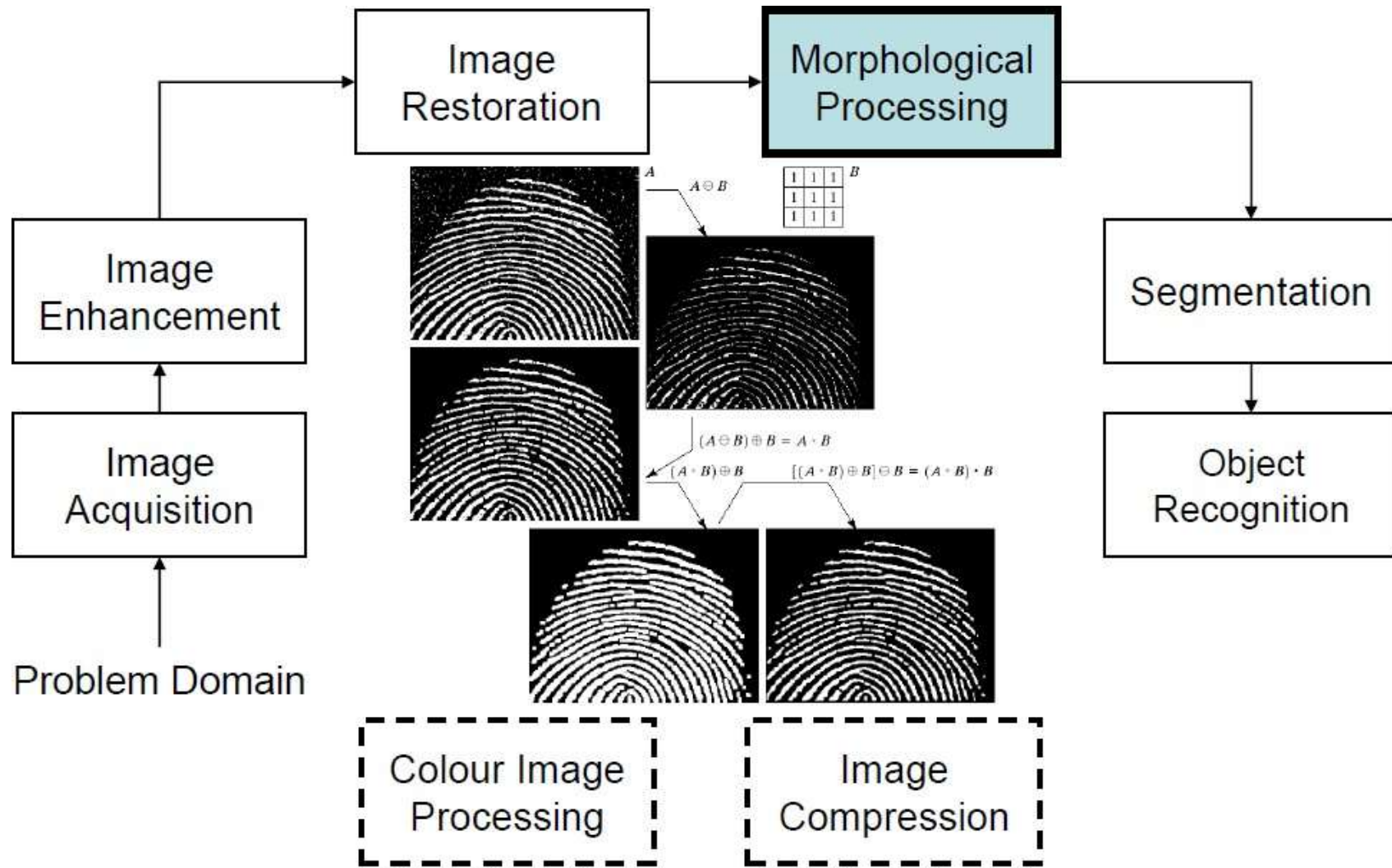
Key stages: Image Enhancement



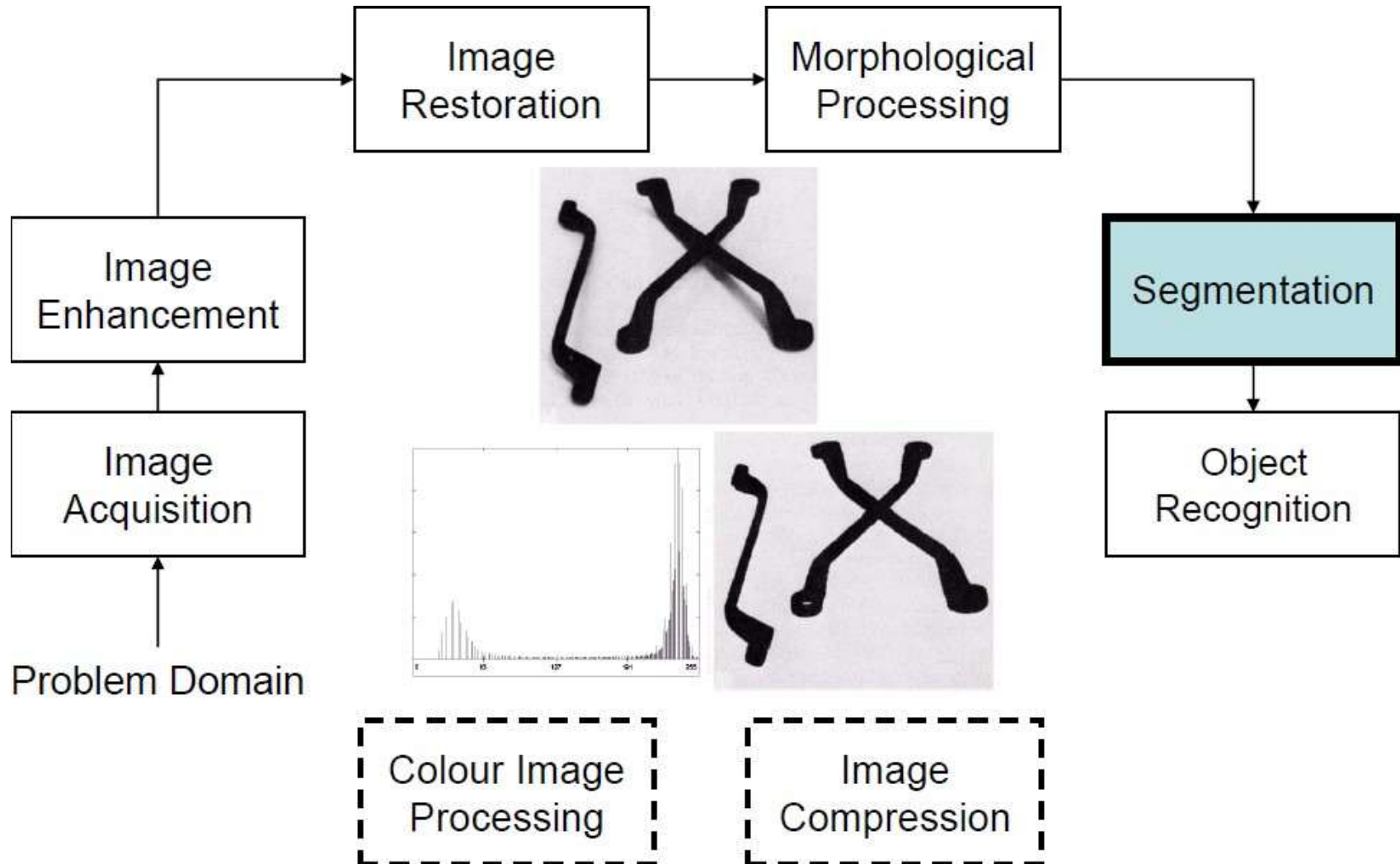
Key stages: Image Restoration



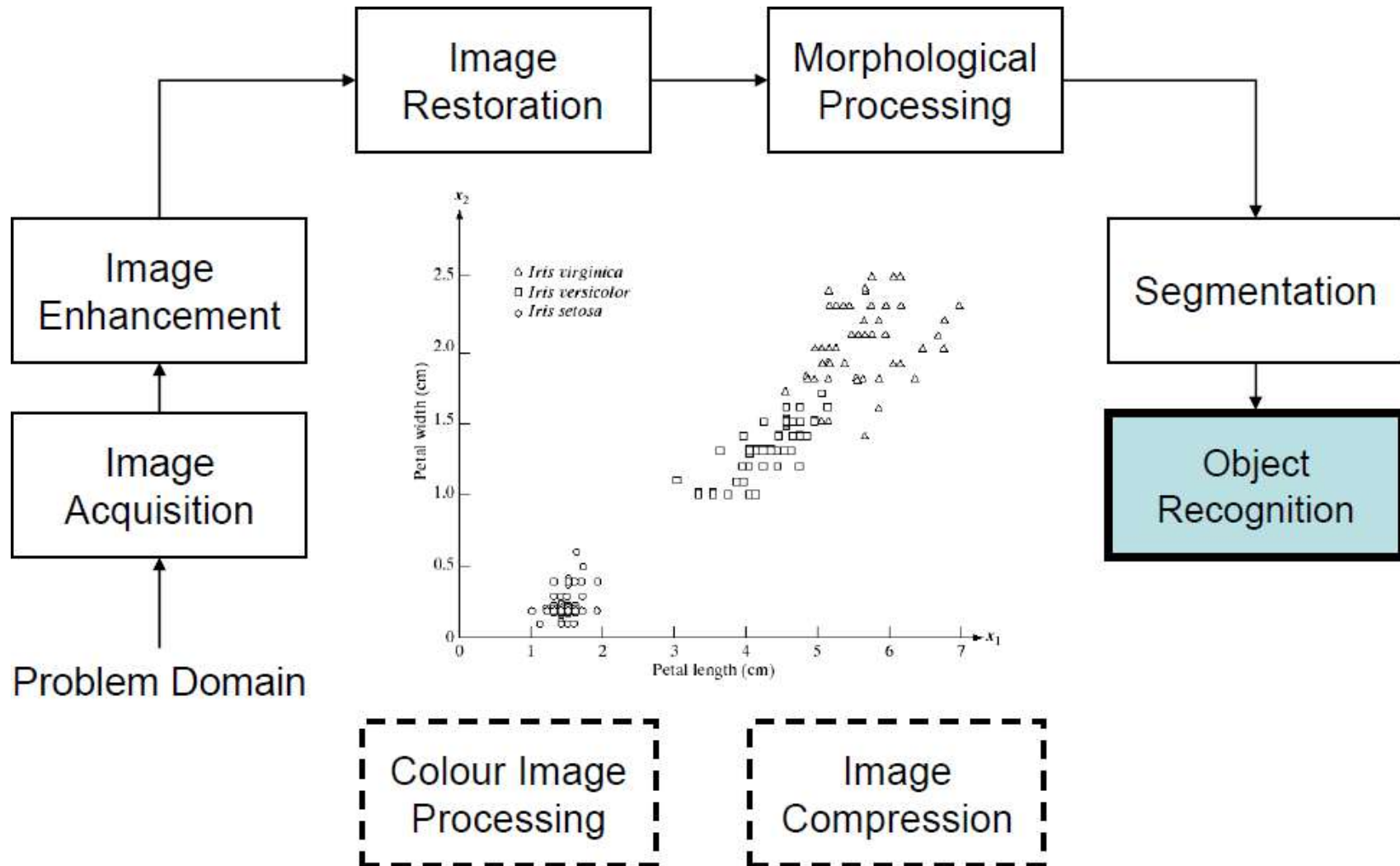
Key stages: Morphological Processing



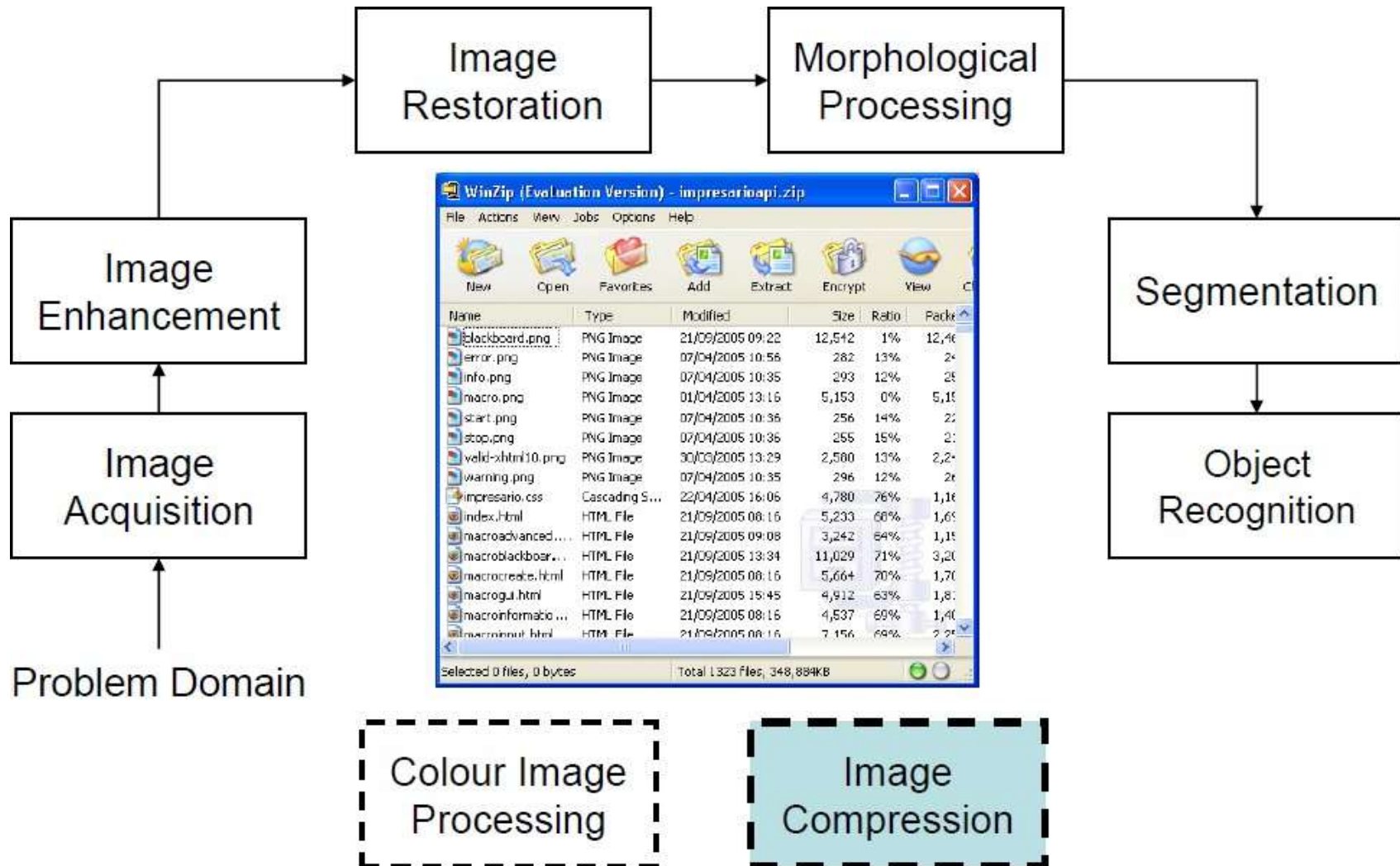
Key stages: Segmentation



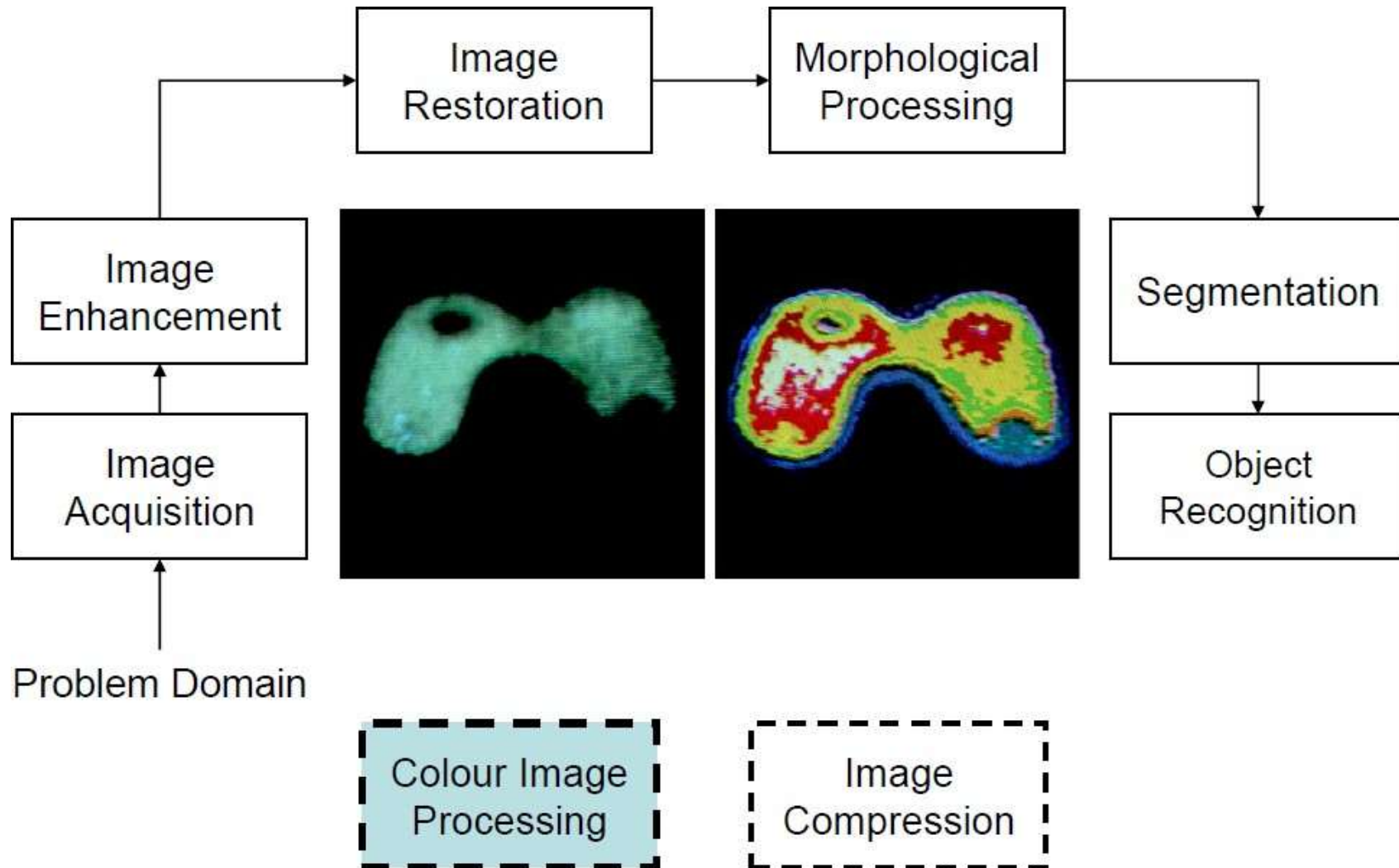
Key stages: Object Recognition



Key stages: Image Compression



Key stages: Colour Image Processing



Course Arrangement

Week 1	Introduction
Week 2	Transforms
Week 3	Morphological Image Processing
Week 4	Image Enhancement in the Spatial Domain
Week 5	Image Enhancement in the Frequency Domain
Week 6	Image Restoration
Week 7	Motion Deblurring using Wiener Filter
Week 8	Color Image Processing — Mid-term Exam (T.B.A.)
Week 9	Image Compression: JPEG as an Example
Week 10	Image Segmentation: Canny Edge Detector
Week 11	Object Recognition: Classical Approaches
Week 12	Face Recognition: Eigen Faces
Week 13	Final Review and Outlook: Deep Learning

- No class on Feb. 18, Apr. 3, Apr. 8, May. 1

Prerequisite Knowledge for This Course

- ▶ Mathematics
 - ▶ Linear Algebra: Vectors and matrices
 - ▶ Probabilities: basic distributions, random variables
 - ▶ Signals and Systems: Transforms, linear systems, filtering, sampling (ELEC 2100)
 - ▶ Digital Signal Processing: ELEC 3100 or ELEC 3200
- ▶ Programming
 - ▶ MATLAB programming experience
- ▶ **This is a mathematics-oriented course!**
- ▶ If you are not familiar with above knowledge:

http://www.imageprocessingplace.com/root_files_V3/tutorials.htm

Teaching Style

- ▶ Textbook is the basic reading material
 - ▶ Digital Image Processing by Gonzalez and Woods , 4th Edition, 2017
<https://www.amazon.com/Digital-Image-Processing-Rafael-Gonzalez/dp/0133356728>
 - ▶ textbooks are reserved in the library
- ▶ Lecture notes cover key concepts
- ▶ Details will be explained on the WHITE BOARD
 - ▶ Your attendance is necessary (although not required) for you to learn the details.
 - ▶ Some details are provided in the additional reading material.
- ▶ Agreement of classroom behavior

Course Grading

- ▶ Homework: 20%
- ▶ Mid-term exam: 30%
- ▶ Final project: 50%
- ▶ No-cheating policy:
 - ▶ Do not copy other people's homework
 - ▶ Do not let other people copy your homework
 - ▶ Both parties will be penalized if found cheating

Additional Information

- ▶ **Course webpage:**

<https://canvas.ust.hk/courses/68288>

- ▶ **Textbooks and tutorials:**

<http://www.imageprocessingplace.com/>

- ▶ **Software-centric books**

- ▶ Gonzalez, Woods, Eddins, Digital Image Processing using Matlab, 2nd edition, Pearson-Prentice-Hall, 2009.
- ▶ Bradski, Kaehler, Learning OpenCV, OReilly Media, 2008.

- ▶ **Comprehensive state-of-the-art**

- ▶ AI Bovik (ed.), The Essential Guide to Image Processing, Academic Press, 2009.

- ▶ **Journals/Conference Proceedings**

- ▶ IEEE Transactions on Image Processing
- ▶ IEEE International Conference on Image Processing (ICIP)
- ▶ IEEE Computer Vision and Pattern Recognition (CVPR)

- ▶ **MATLAB:**

Mathworks webpage

www.mathworks.com/academia/student_center/tutorials/launchpad.html

Access to the Course Webpage

- ▶ If you enroll to the course:
 - ▶ Log in at <https://canvas.ust.hk/courses/68288> using your ITSC ID and password
- ▶ If you are auditing this course, please let me know.
- ▶ If you want to get the score for auditing, please finish all homework, exam and project.

Questions?

► Part II Fundamentals

Outline

Review

Human Visual Perception

Image formation

Image Sensing and Acquisition

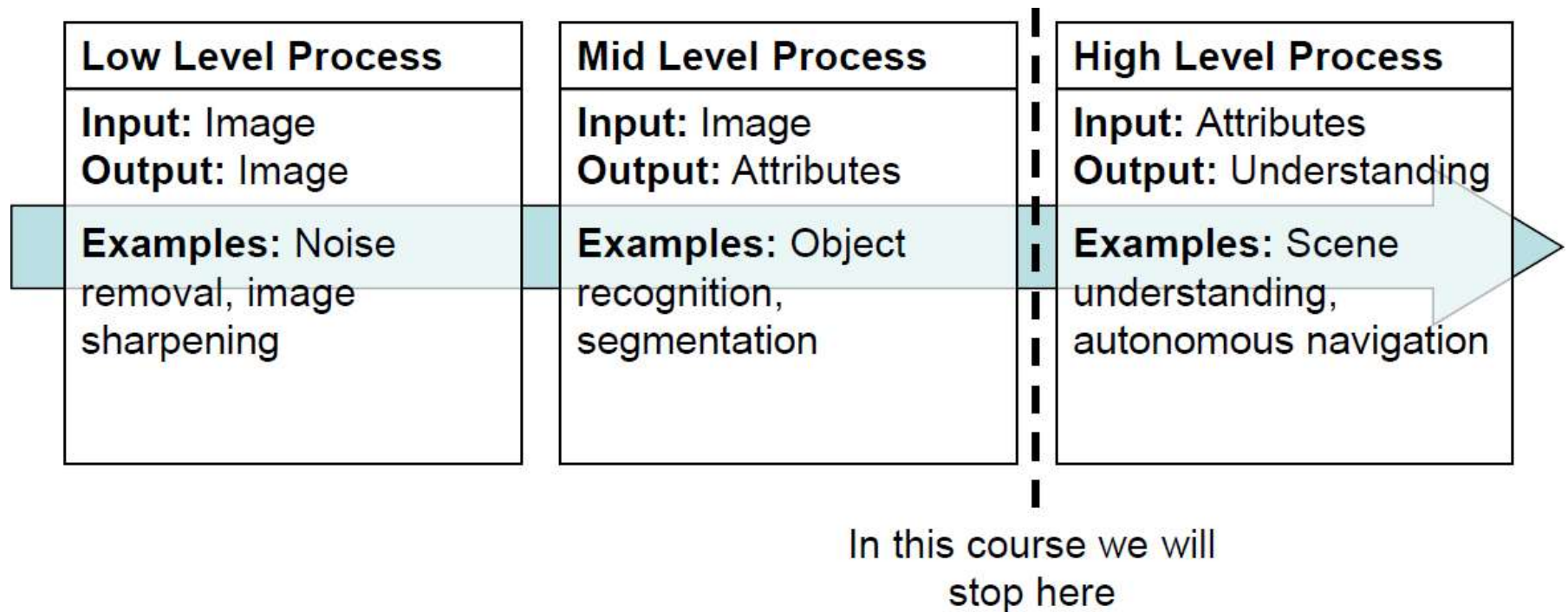
Sampling and Quantization

Shrinking and Zooming

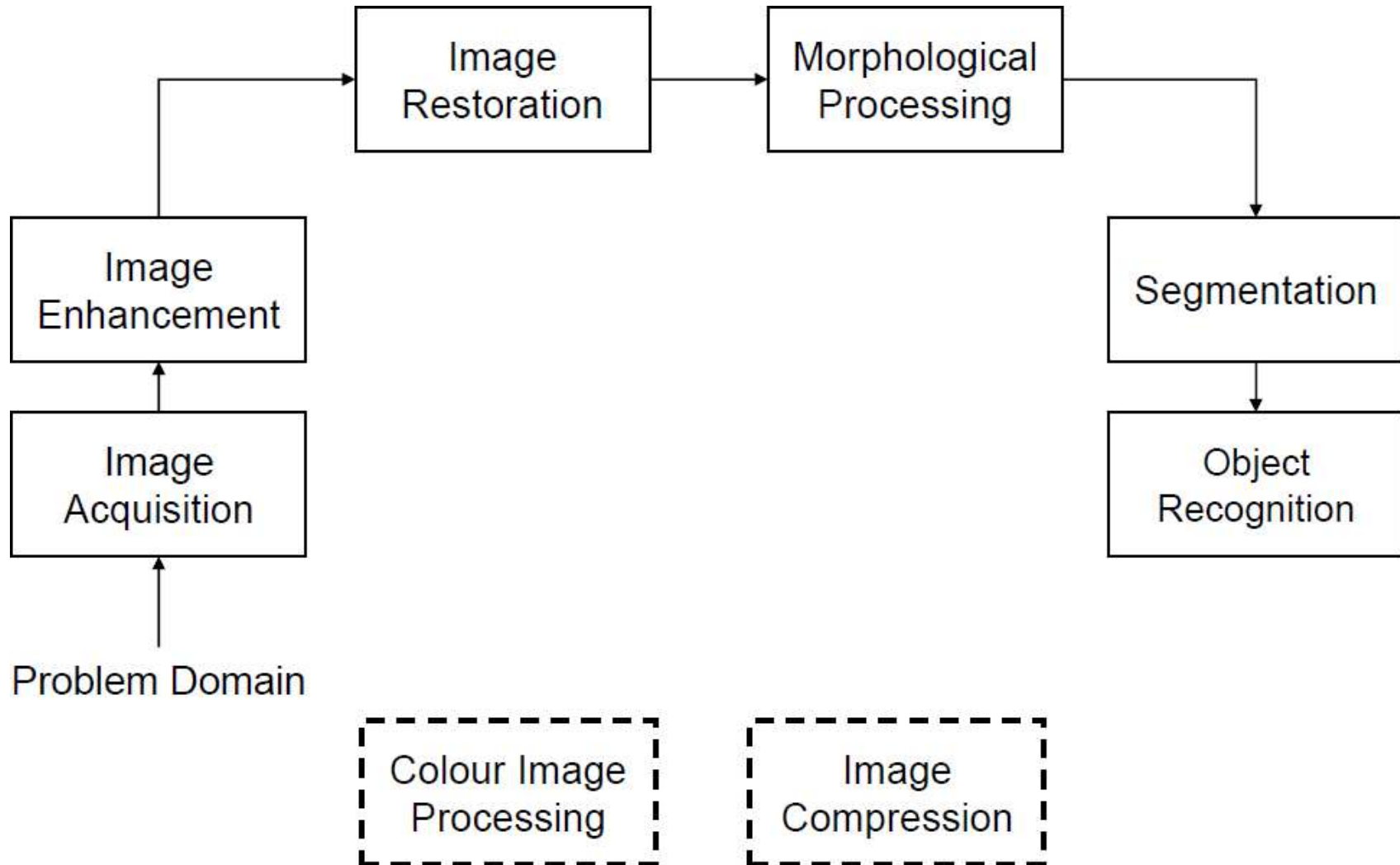
Basic Mathematical Tools

Review: 3-level process

- ▶ The continuum from image processing to computer vision can be broken up into low-, mid- and high-level processes



Review: Key stages



Review: Two Principles

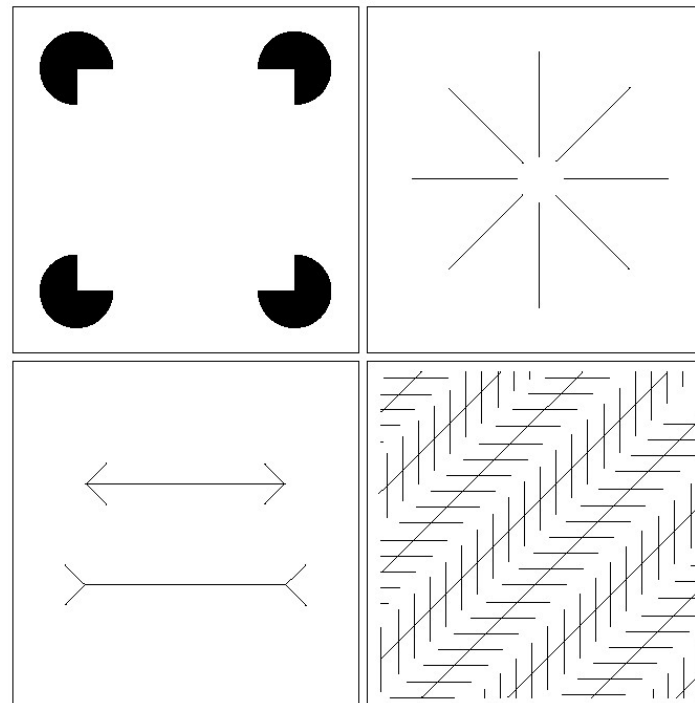
- ▶ Focuses on two major tasks
 - ▶ Improvement of pictorial information for human interpretation
 - ▶ Processing of image data for storage, transmission and representation for autonomous machine perception

Human Visual Perception

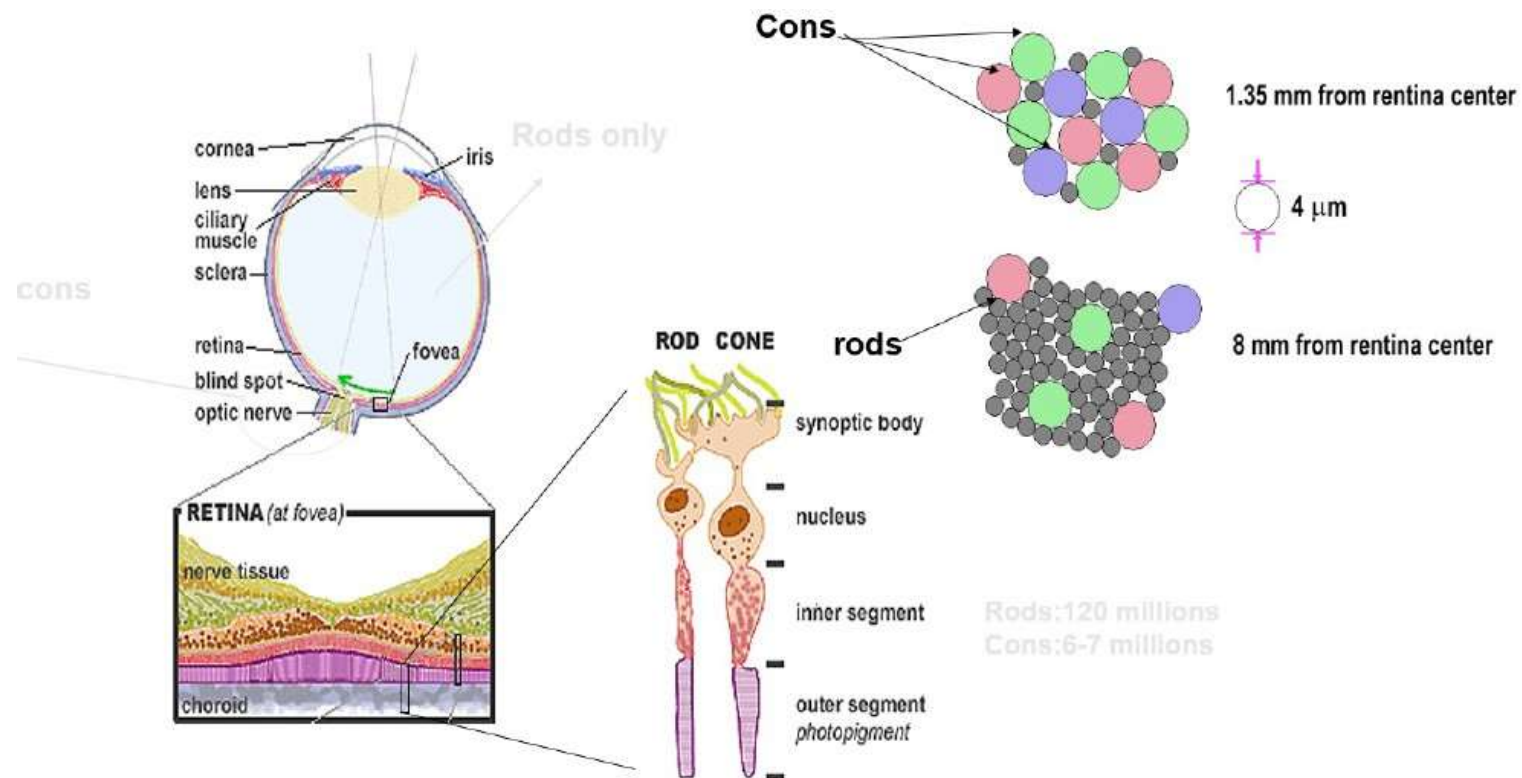
- ▶ Why is it important?
 - ▶ Visual judgment is often used in DIP results (Although not always so)
 - ▶ DIP algorithms sometimes mimic human perception principles:
 - ▶ Geometry model in camera design
 - ▶ biological vision
 - ▶ AI
- ▶ Visual perception has Illusions

a b
c d

FIGURE 2.9 Some well-known optical illusions.



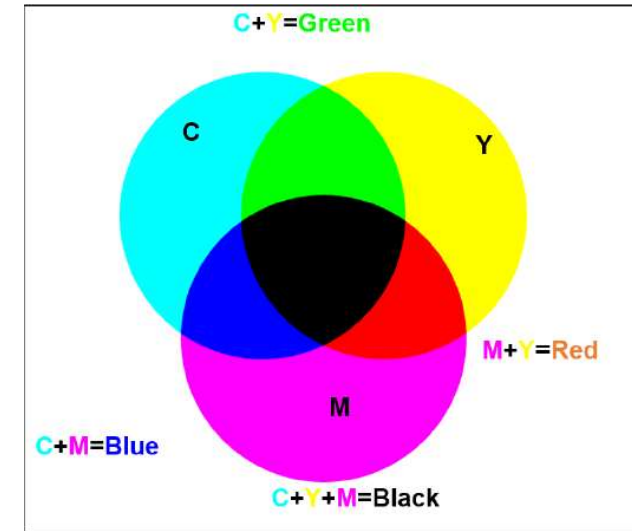
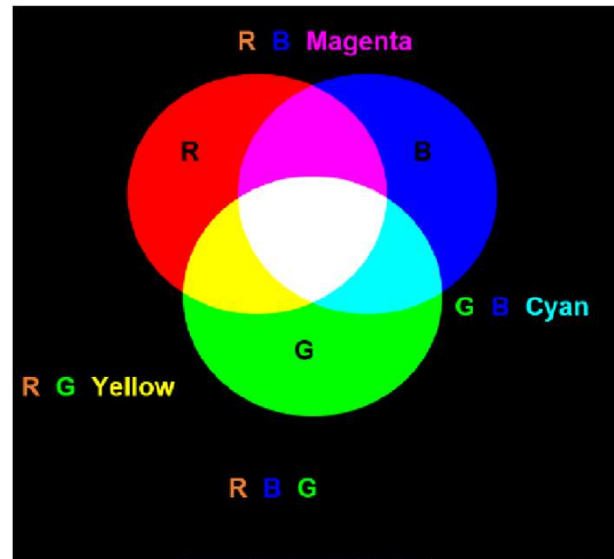
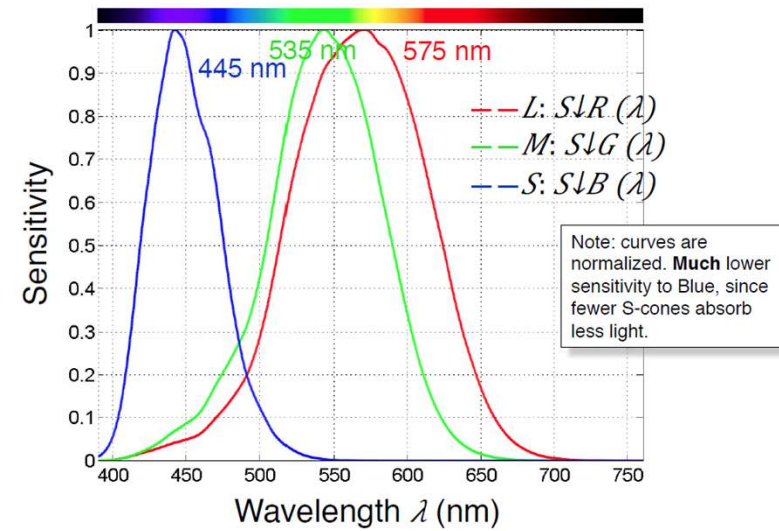
HVS



*Cones: luminance (bright) and color

*Rods: luminance only, but 10,000 times sensitive to cones

Color Space



Main Contents

Review

Human Visual Perception

Image formation

Image Sensing and Acquisition

Sampling and Quantization

Shrinking and Zooming

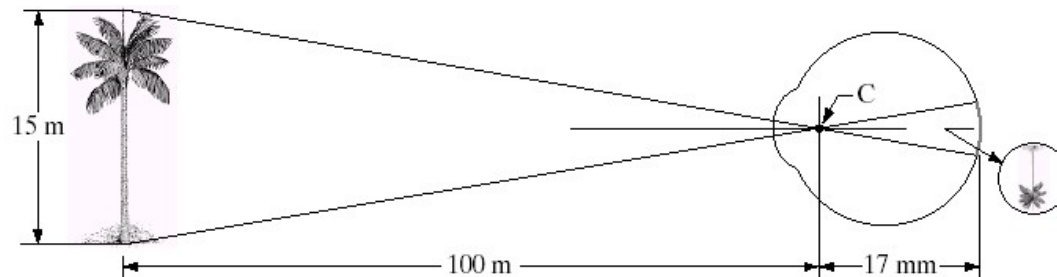
Basic Mathematical Tools

Image Sensing and Acquisition

► Human perception

FIGURE 2.3

Graphical representation of the eye looking at a palm tree. Point *C* is the optical center of the lens.



► Pin-hole camera model: simple projection

Imaging with Sensor Arrays

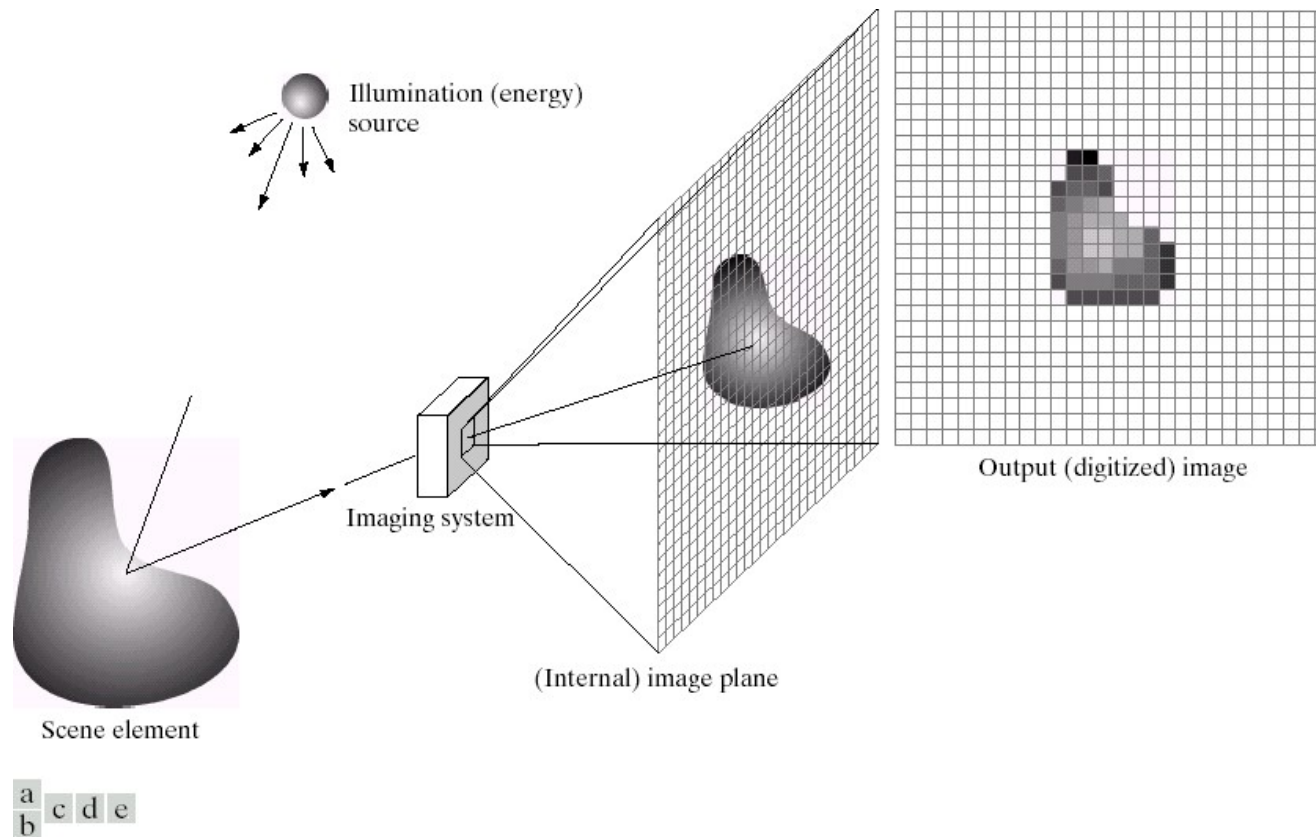


FIGURE 2.15 An example of the digital image acquisition process. (a) Energy (“illumination”) source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

Analog to digital conversion

- ▶ Sampling: Sampling frequency should satisfy Nyquist Criterion
- ▶ Quantization

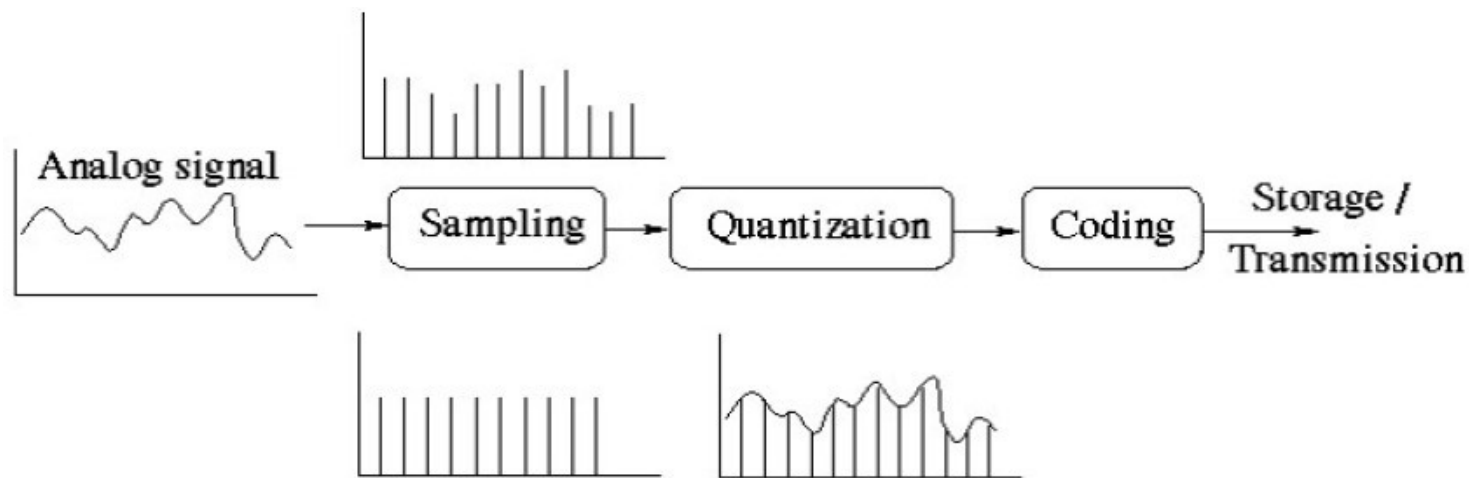
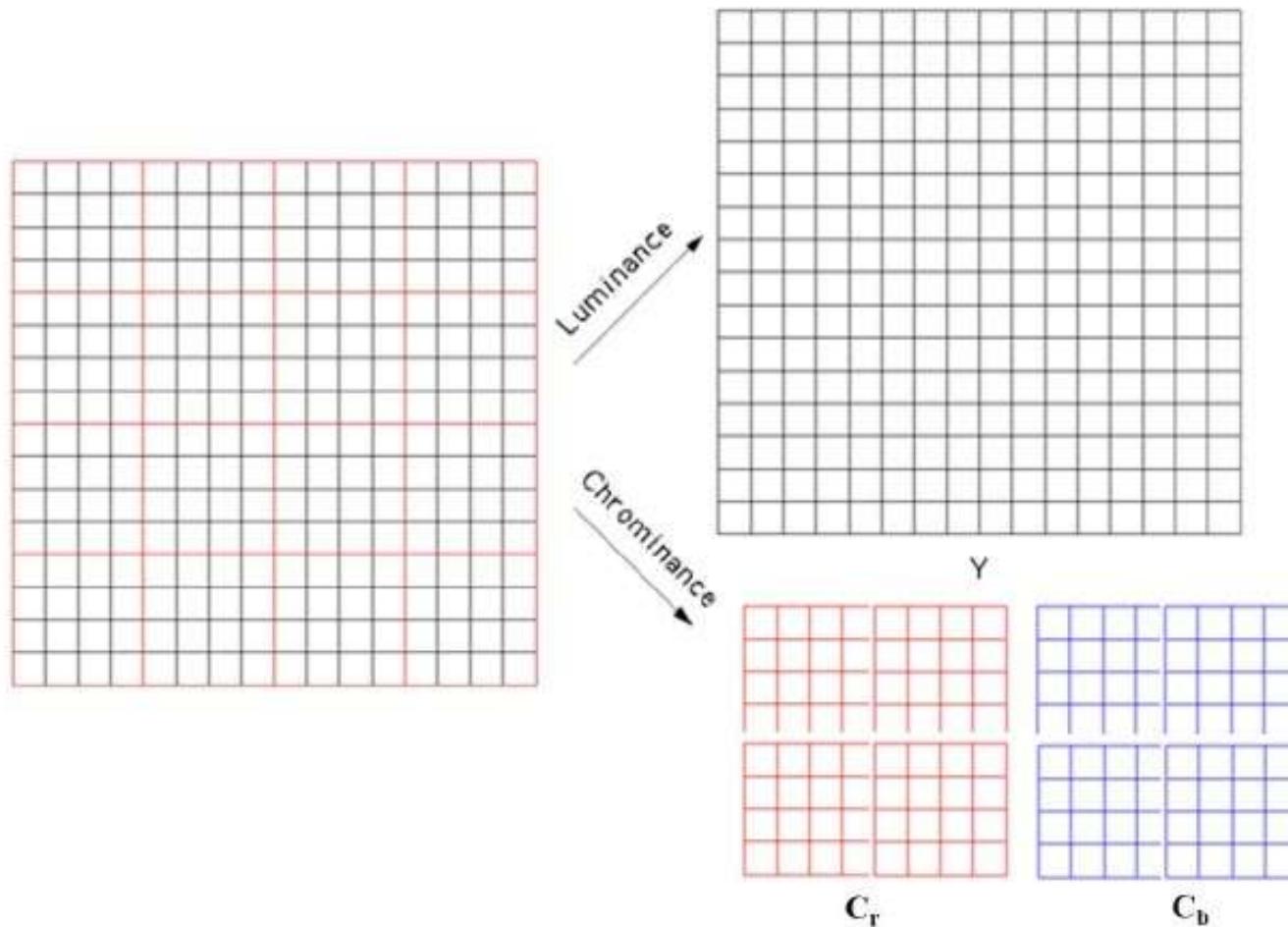


Image Sampling



Main Contents

Review

Human Visual Perception

Image formation

Image Sensing and Acquisition

Sampling and Quantization

Shrinking and Zooming

Basic Mathematical Tools

Sampling and Quantization of Digital Images Using CCD Cameras

- ▶ In CCD camera, sampling and quantization are key elements
- ▶ **Sampling:** Picture sampled by CCD chip grid
- ▶ **Quantization:** Each CCD element uses fixed number of bits to represent image values
⇒ We represent an image as a matrix of numbers
⇒ MATLAB is the ideal tool to process image

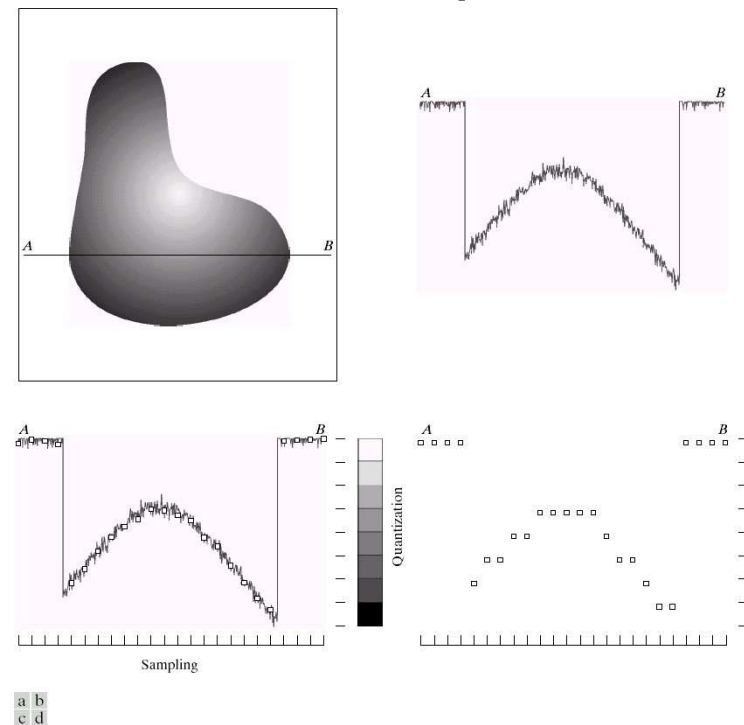


FIGURE 2.16 Generating a digital image. (a) Continuous image. (b) A scan line from A to B in the continuous image, used to illustrate the concepts of sampling and quantization. (c) Sampling and quantization. (d) Digital scan line.

Sampling and Quantization

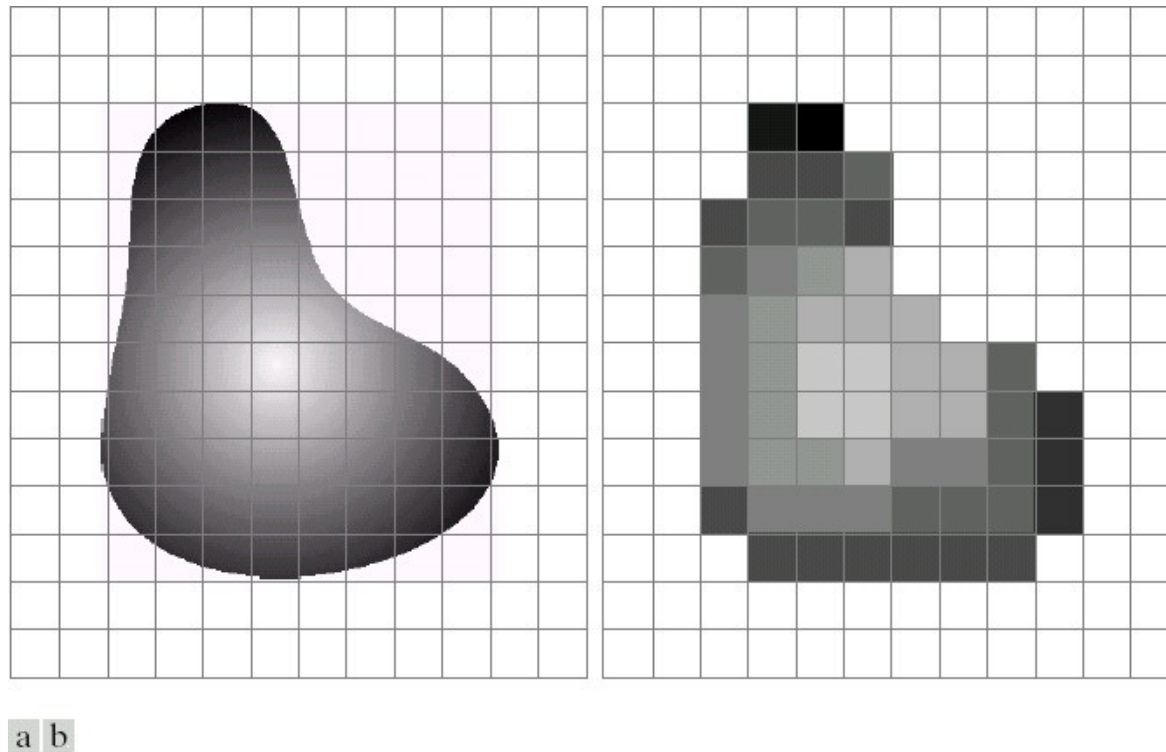
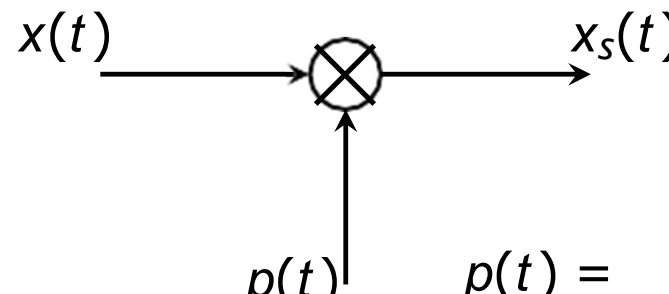


FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

How is Sampling Done?

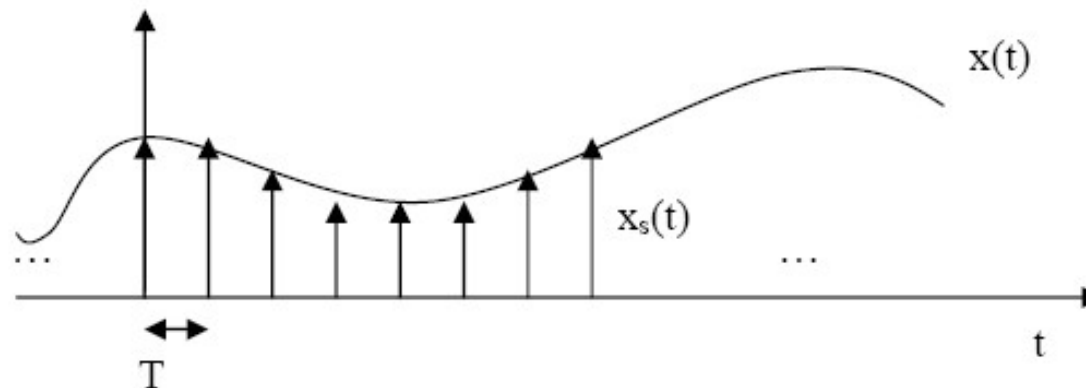
- By applying the Dirac comb function:



A block diagram showing the sampling process. An input signal $x(t)$ enters a multiplier block (a circle with an 'X'). A sampling pulse train $p(t)$ enters the same multiplier block from below. The output of the multiplier is the sampled signal $x_s(t)$.

$$p(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$$

- 1-D case:



Dirac Function:

- ▶ Definition:

$$\delta(x) = \begin{cases} +\infty, & x = 0 \\ 0, & x \neq 0 \end{cases} \quad \text{and} \quad \int_{-\infty}^{+\infty} \delta(x) dx = 1$$

- ▶ Important Properties:

- ▶ Sifting property: $\int_{-\infty}^{+\infty} f(x) \delta(x - a) dx = f(a)$
- ▶ Convolution property: $f(t) * \delta(t - T) = f(t - T)$
- ▶ FT of Dirac:
 - ▶ $1 \rightarrow \delta(u)$
 - ▶ $\delta(t) \rightarrow 1$

Dirac Comb Function (Shah Function)

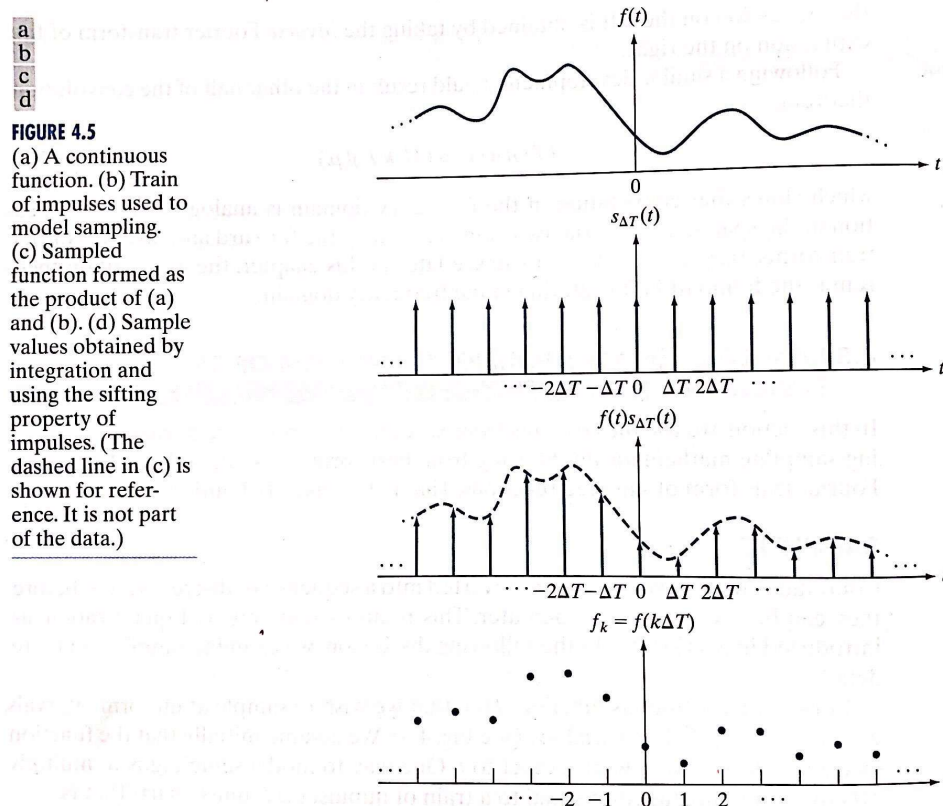
- ▶ Definition:

$$\Delta_T(t) \stackrel{\text{def}}{=} \sum_{k=-\infty}^{\infty} \delta(t - kT).$$

- ▶ FT of Dirac Comb:

$$\sum_{n=-\infty}^{\infty} \delta(t - nT) \stackrel{\text{F}}{\Longleftrightarrow} \frac{1}{T} \sum_{k=-\infty}^{\infty} \delta\left(f - \frac{k}{T}\right) = \sum_{n=-\infty}^{\infty} e^{-i2\pi f n T}.$$

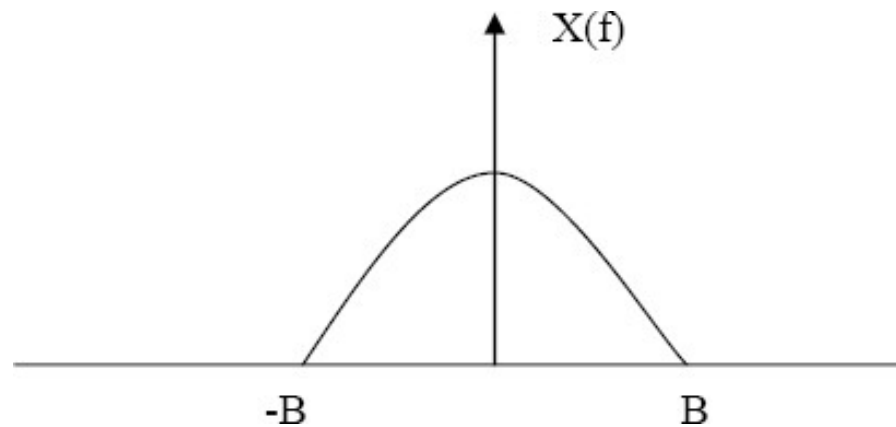
Effect of Sampling in the Spatial Domain



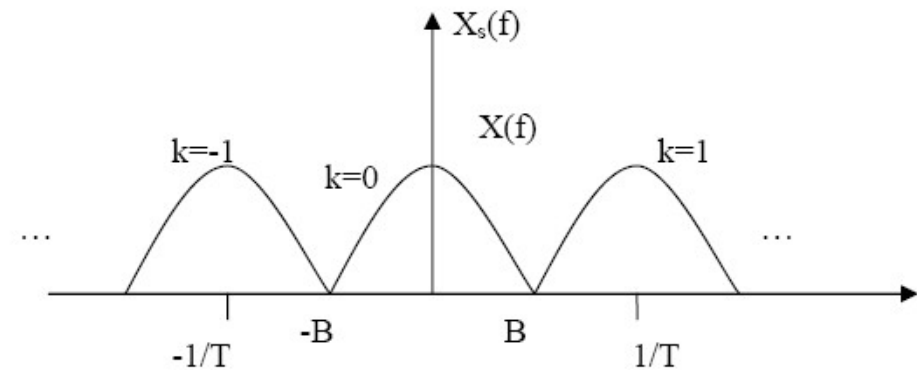
- ▶ Sampling = Original continuous signal $\times$ the Dirac comb function
- ▶ Question: Why is it not convolution ?

Effect of Sampling in the Frequency Domain

- ▶ Spectrum of original signal
Before Sampling:

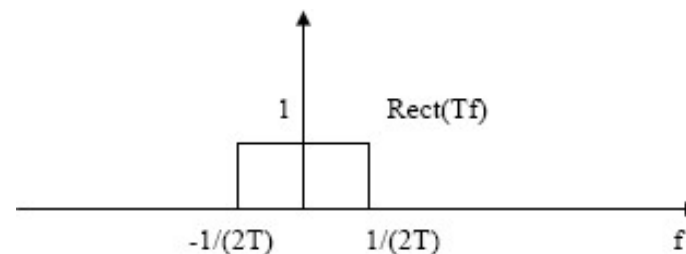


After Sampling:



Effect of Sampling in the Frequency Domain

- ▶ Key element in sampling: sampling rate
- ▶ Nyquist rate: the minimal rate guaranteeing no overlapping of neighboring spectra.
 - ⇒ Shannon's famous sampling theorem (Reading material)
- ▶ Obtain original signal from sampled version:
- ▶ Low-pass filtering
- ▶ Inverse Fourier transform



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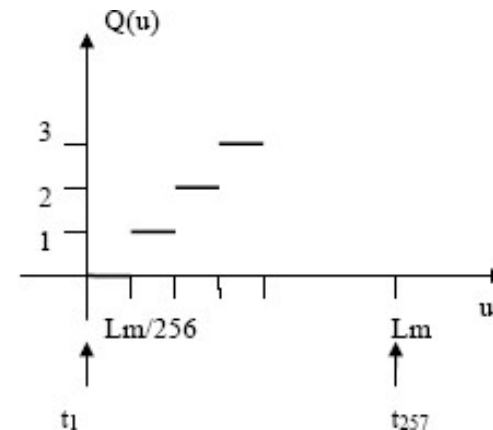
Quantization

- ▶ Why? Only limited bits (precision) available
- ▶ How? See the following example:
 - ▶ 8 bits (256 levels) to represent all possible values
 - ▶ A linear quantization scheme:

$$Q(u) = k \quad \text{if} \quad t_k \leq u < t_{k+1}$$

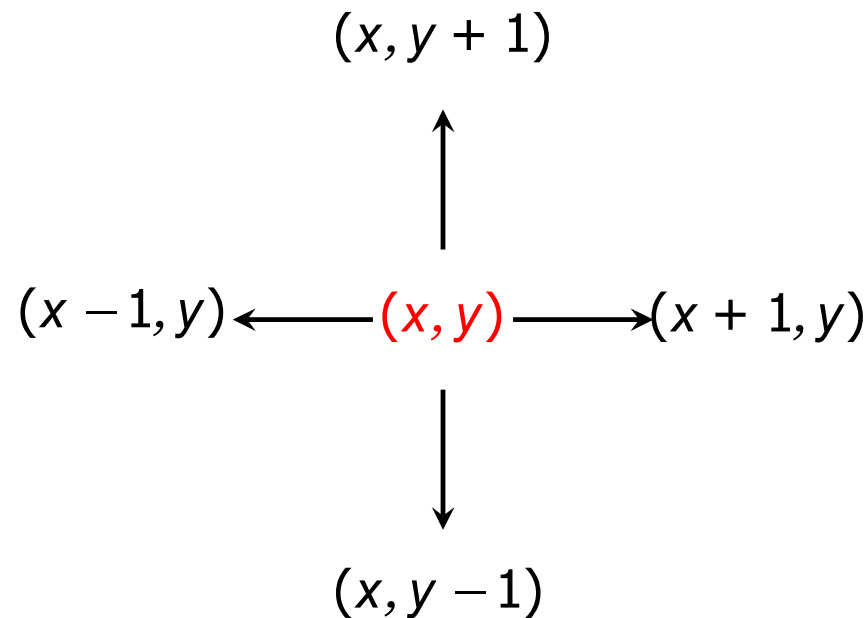
- ▶ Different quantization schemes based on:

1. Value Range
2. Value Distribution



Spatial Relations in Images

- ▶ Element unit: pixel with coordinates (x, y)
- ▶ **Four** neighbors of pixel (x, y)



Spatial Relations in Images

- ▶ Element unit: pixel with coordinates (x, y)
- ▶ **Eight** neighbors of pixel (x, y)

$$(x - 1, y + 1) \quad (x, y + 1) \quad (x + 1, y + 1)$$

$$(x - 1, y) \quad (x, y) \quad (x + 1, y)$$

$$(x - 1, y - 1) \quad (x, y - 1) \quad (x + 1, y - 1)$$

Distance Measures

- ▶ A function $D(p, q)$ is a distance measure if:
 - ▶ $D(p, q) \geq 0$.
 - ▶ $D(p, q) = D(q, p)$.
 - ▶ $D(p, z) \leq D(p, q) + D(q, z)$.
- ▶ Common distance measures:
 - ▶ Euclidean distance (L_2): $D_e(p, q) = \sqrt{(x - s)^2 + (y - t)^2}$.
 - ▶ City-block distance (L_1): $D_4(p, q) = |x - s| + |y - t|$.
 - ▶ Chessboard distance (L_∞): $D_\infty(p, q) = \max(|x - s|, |y - t|)$.
- ▶ Distance measure is critical in spatial filtering

Main Contents

Review

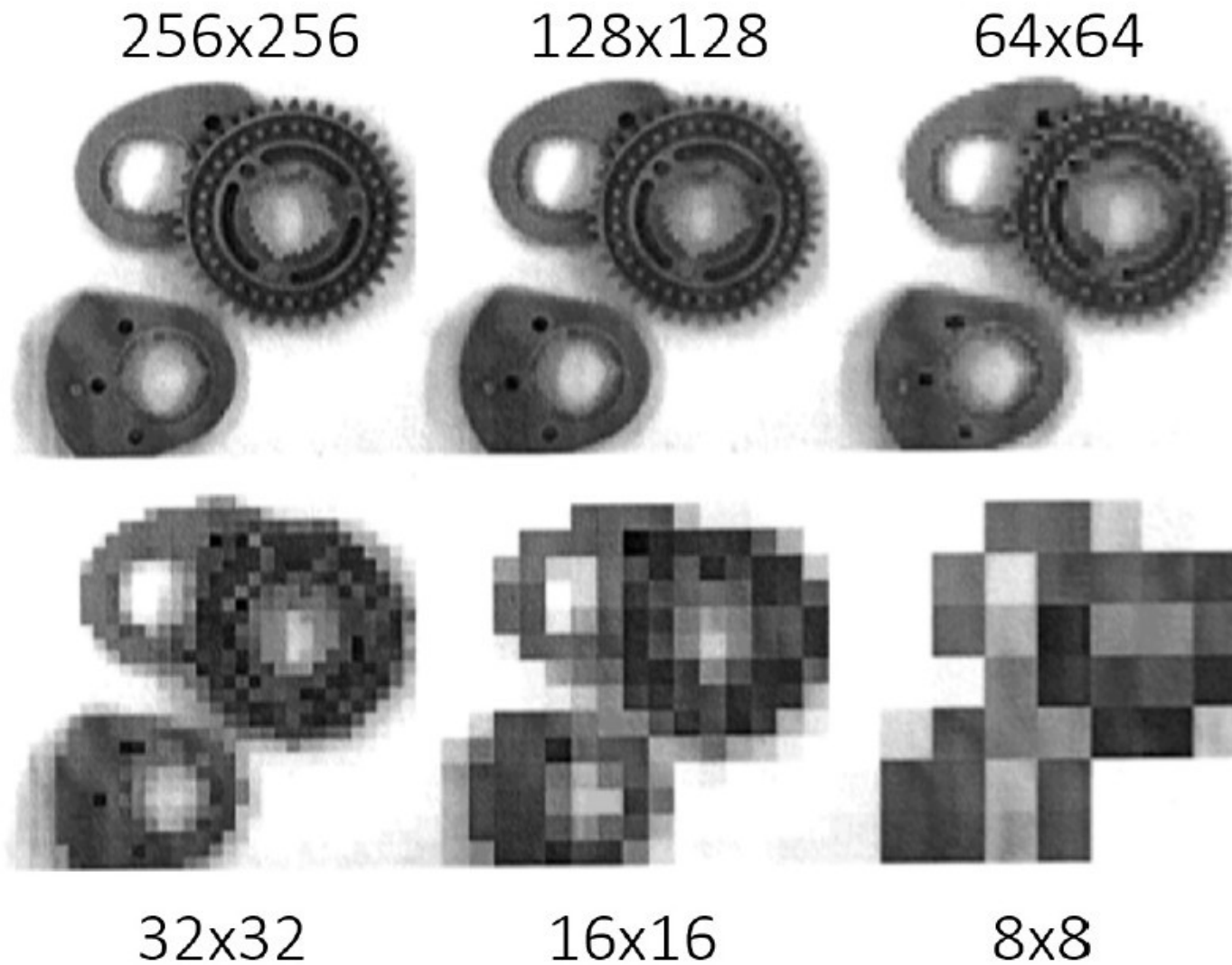
Human Visual Perception

Image formation

Image Sensing and Acquisition
Sampling and Quantization
Shrinking and Zooming

Basic Mathematical Tools

Image Sampling



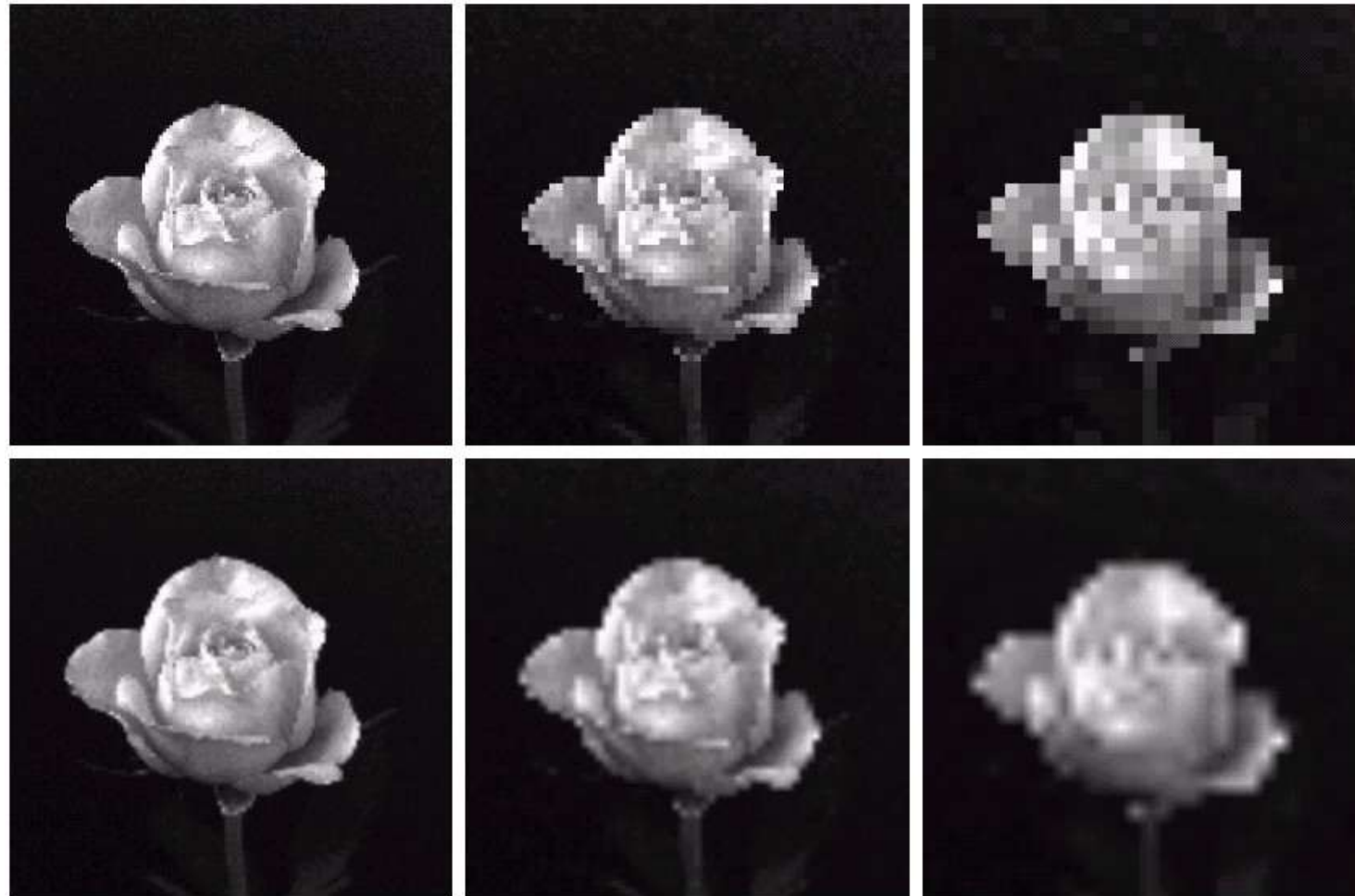
Shrinking and Zooming

- ▶ In analog images (such as films): resolution depends on the density of light-sensitive materials
- ▶ In digital images: resolution depends on the sampling rate
- ▶ Shrinking: Delete every other row and column
 - ⇒ increase sampling interval
 - ⇒ Lose details (high frequency parts)
- ▶ Zooming: Increase the number of rows and columns

Zooming Details

- ▶ Simple example: $M \times N$ to $2M \times 2N$ (double grid)
- ▶ How to assign values to the new matrix?
 - ▶ Interpolation methods often used
 - ▶ Original value at every-other row/column (1, 3, ...)
 - ▶ Nearest neighbor interpolation: Duplicate value at the rest grids (2, 4, ...)
 - ▶ Bilinear interpolation: $V(x, y) = ax + by + cxy + d$.
 - ▶ ...
- ▶ Smoothing often used to make image looking better (involves spatial filtering and distance measure)
- ▶ Quantitative measure is a better criterion in judging the zoomed image quality

Examples



a b c
d e f

FIGURE 2.25 Top row: images zoomed from 128×128 , 64×64 , and 32×32 pixels to 1024×1024 pixels, using nearest neighbor gray-level interpolation. Bottom row: same sequence, but using bilinear interpolation.

Basic Mathematical Tools

- ▶ Element-wise vs. Matrix Operation
- ▶ Linear vs. Nonlinear Operation
- ▶ Arithmetic Operation
- ▶ Set and Logical Operation
- ▶ Spatial Operation
- ▶ Probability and Random Variables
- ▶ Vector and Matrix Operation

Element-wise vs. Matrix Operation

- ▶ Element-wise product

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} \\ a_{21}b_{21} & a_{22}b_{22} \end{bmatrix}$$

- ▶ Matrix product

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

Linear vs. Nonlinear Operation

Consider a general operator, H , that produces an output image, $g(x, y)$, for a given input image, $f(x, y)$:

$$H[f(x, y)] = g(x, y)$$

H is said to be a linear operator if

$$\begin{aligned} H[a_i f_i(x, y) + a_j f_j(x, y)] &= a_i H[f_i(x, y)] + a_j H[f_j(x, y)] \\ &= a_i g_i(x, y) + a_j g_j(x, y) \end{aligned}$$

Arithmetic Operation

- Four arithmetic operations

$$s(x, y) = f(x, y) + g(x, y)$$

$$d(x, y) = f(x, y) - g(x, y)$$

$$p(x, y) = f(x, y) \times g(x, y)$$

$$v(x, y) = f(x, y) \div g(x, y)$$

- Example: Noise reduction by averaging over noisy images.

Set and Logical Operation

- ▶ Basic set operations: subset, union, intersection, complement
...
- ▶ Logical operations: AND, OR, NOT, XOR...
- ▶ Applications: Morphological processing

Spatial Operation

- ▶ Single pixel operation.
Example: find the complement of image
- ▶ Neighborhood Operations.
Example: Local averaging
- ▶ Geometric Transformation
 - ▶ Spatial transformation of coordinates
 - ▶ Intensity interpolation
- ▶ Image Registration

Transformation Name	Affine Matrix, T	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v$ $y = w$	
Scaling	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = c_x v$ $y = c_y w$	
Rotation	$\begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v \cos \theta - w \sin \theta$ $y = v \sin \theta + w \cos \theta$	
Translation	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ t_x & t_y & 1 \end{bmatrix}$	$x = v + t_x$ $y = w + t_y$	
Shear (vertical)	$\begin{bmatrix} 1 & 0 & 0 \\ s_v & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v + s_v w$ $y = w$	
Shear (horizontal)	$\begin{bmatrix} 1 & s_h & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v$ $y = s_h v + w$	

Probability and Random Variables

- ▶ Sample space, events and probability
- ▶ Conditional probability, Bayes' rule
- ▶ Independence, the law of total probability
- ▶ Random variables, Probability Mass Function, Probability Density Function
- ▶ Mean, variance, and higher order moments

Introduction of Matrix Algebra

- ▶ Indexing in matrices and images. Try this example in Matlab:

```
a = meshgrid([1:10]);  
a(1,2) = 100;  
figure; imagesc(a); grid on; axis image;  
figure; imagesc(a); grid on; axis xy;
```

Basic Rules and Definitions in Matrix Algebra

► Let

$$A = \{a[m, n]\} = \begin{bmatrix} a[1, 1] & a[1, 2] & \cdots & a[1, N] \\ a[2, 1] & \ddots & & \vdots \\ \vdots & & & \vdots \\ a[M, 1] & a[M, 2] & \cdots & a[M, N] \end{bmatrix}$$

► Transpose:

$$A^T = \{a[n, m]\},$$

$$\text{e.g. } A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Basic Rules and Definitions in Matrix Algebra

- Addition and multiplication:

$$C = A + B = \{ a[m, n] + b[m, n] \}$$

$$C = A \cdot B = \left\{ \sum_{k=1}^k a[m, k] b[k, n] \right\} .$$

- Symmetric matrix

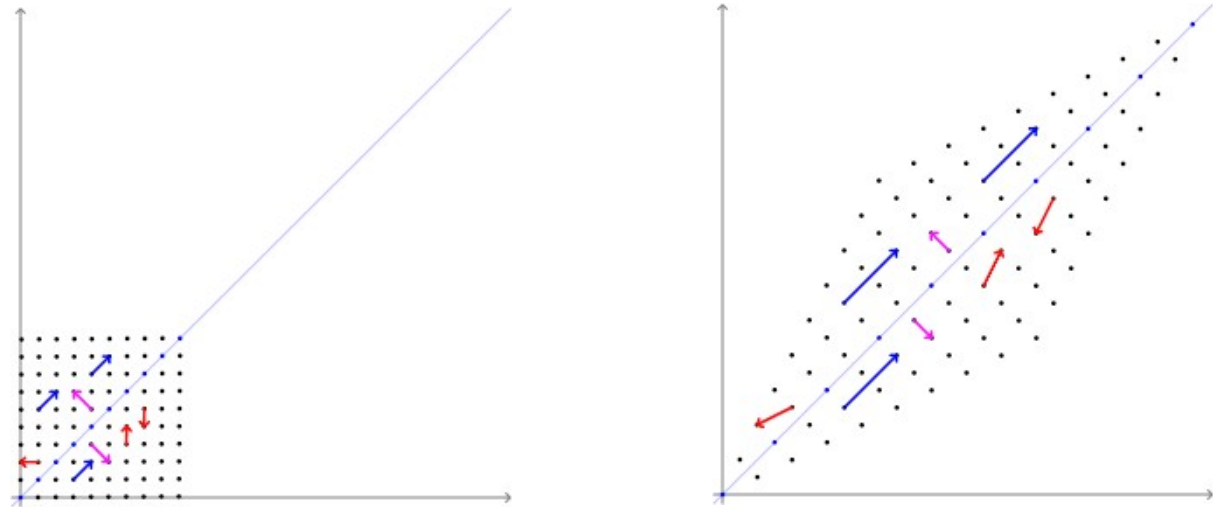
$$A^T = A, \quad \text{Symmetric} \quad \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix},$$

Basic Rules and Definitions in Matrix Algebra

- ▶ Eigenvalues and Eigenvectors:
For a matrix P , if there is a scalar λ and a vector v such that:
 $Pv = \lambda v$, then λ : eigenvalue (characteristic value) and v :
eigenvector
 $\Rightarrow$ Eigenvectors are linearly independent if they correspond to
distinct eigenvalues.

More on Eigenvalues and Eigenvectors

- ▶ Illustration



- ▶ How to calculate λ and v ?

$$Pv = \lambda v \Rightarrow |P - \lambda I| = 0.$$

- ▶ How to understand λ and v ?

- ▶ v : basis of the vector space.
- ▶ λ : weighting coefficient of the vector.

Reading Assignment

- ▶ Chapter 1 and 2 of the textbook